

Peter Richtárik: Curriculum Vitae

1. CONTACT DETAILS

Address: Office 3145, Building 12, 4700 KAUST, Thuwal 23955-6900, Kingdom of Saudi Arabia
Email: peter.richtarik@kaust.edu.sa
Website: <https://richtarik.org>

2. RESEARCHER IDs

dblp: <https://dblp.org/pid/62/8001.html>
orcid: <https://orcid.org/0000-0003-4380-5848>
Web of Science Researcher ID: O-5797-2018

3. RESEARCH INTERESTS

- ◇ machine learning, federated learning, empirical risk minimization
- ◇ big data optimization, convex and non-convex optimization; 0th, 1st, and 2nd order optimization methods
- ◇ randomized algorithms, randomized coordinate descent, stochastic gradient descent, variance reduction
- ◇ randomized numerical linear algebra
- ◇ parallel and distributed computing, supercomputing, gradient compression

4. ACADEMIC POSITIONS

2019–now **Professor**, Computer Science, King Abdullah University of Science and Technology (KAUST), Kingdom of Saudi Arabia
2022–2023 **Adjunct Professor**¹, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, United Arab Emirates
2017–2019 **Visiting Professor**, Moscow Institute of Physics and Technology, Russia
2017–2019 **Associate Professor**, Computer Science, KAUST, Kingdom of Saudi Arabia
2016–2019 **Associate Professor (Reader)**, Mathematics, University of Edinburgh
2013 **Invited Visiting Scientist**, Simons Institute for the Theory of Computing, UC Berkeley
2009–2016 **Assistant Professor (Lecturer)**, School of Mathematics, University of Edinburgh
2007–2009 **Postdoctoral Fellow**, Center for Operations Research and Econometrics and Department of Mathematical Engineering, Catholic University of Louvain, Belgium (host: Yu. Nesterov)

5. EDUCATION

2007 **PhD, Operations Research, Cornell University**
2006 MS, Operations Research, Cornell University
2001 Mgr, Mathematics, Comenius University, Slovakia, 100% academic grades, ranked #1
2001 Bc, Management, Comenius University, Slovakia, 100% academic grades, ranked #1
2000 Bc, Mathematics, Comenius University, Slovakia, 100% academic grades, ranked #1

¹This position allowed me to supervise Dr. Sarit Khirirat – a postdoc located at MBZUAI.

6. AWARDS & RECOGNITIONS

Awards explicitly addressed to my students, postdocs or coauthors for a talk, poster or paper based on joint research with me are listed in the section “8.4 My Team: Awards and Recognitions”.

2026	2023 Charles Broyden Prize² (for paper [94])
2025	Spotlight Paper at ICLR 2025 (for paper [244])
2025	AMA Distinguished Speaker , The Hong Kong Polytechnic University
2024	Top 8–14 Author at NeurIPS 2024 in terms of the number of papers accepted to the conference ³
2024	2022 Charles Broyden Prize⁴ (for paper [139])
2024	Oral Paper at NeurIPS 2024 (0.4% acceptance rate ⁵ ; paper [249])
2024	Spotlight Paper at NeurIPS 2024 (for paper [243])
2024	Oral Paper at ICML 2024 (1.5% acceptance rate; paper [231])
2023	#1 in Asia and #3 in Europe in Machine Learning according to CSRankings.org⁶
2023	Research.com Computer Science in Saudi Arabia Leader Award
2023	Research.com Mathematics in Saudi Arabia Leader Award
2023	Oral Paper at ICLR 2023 (for paper [184])
2022	Top 10–14 Author at NeurIPS 2022 in terms of the number of papers accepted to the conference ⁷
2022	Research.com Rising Star of Science ⁸ , global rank 214 among all fields of science
2022	Spotlight Paper at ICLR 2022 (for paper [156])
2021	Oral Paper at NeurIPS 2021 (less than 1% acceptance rate; paper [167])
2021	2020 COAP Best Paper Award⁹ (for paper [65])
2021	One of the 10 Most Cited Articles Published in SIMAX Since 2019¹⁰ for paper [56]
2020	Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning for paper [135]
2020	Top 30–50 author at ICML 2020 (in number of papers accepted)
2020	1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]
2020	3rd Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]
2020	3rd Most Downloaded Paper in “SIAM J. on Optimization” for paper [57]
2020	4th Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]
2019	1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]
2019	4th Most Downloaded Paper in “SIAM J. on Optimization” for paper [57]
2019	5th Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]
2019	10th Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]
2019	Interviewed by Robin.ly for their “Leaders in AI” platform at NeurIPS 2019¹¹

²The prize was announced in 2026.

³<https://papercopilot.com/paper-list/neurips-paper-list/neurips-2024-paper-list/>

⁴The prize was announced in 2024.

⁵61 papers out of more than 15,000.

⁶As all rankings, these rankings have to be taken with a huge grain of salt. According to CS Rankings in the 5 year period 2019–2023, I rank #1 in Machine Learning in all of Asia: 1) Peter Richtárik at KAUST (14), 2) Sung Ju Hwang at KAIST (13.6), 3) Jinwoo Shin at KAIST (13.2), 4) Jun Zhu at Tsinghua (12.3), 5) Masashi Sugiyama at Tokyo (12), 6) Junchi Yan at Shanghai Jiao Tong (10.4) 7) Taiji Suzuki at Tokyo (9.6), 8) Kun Zhang at MBZUAI (9.3). If KAUST was in Europe, I would rank #3 in Europe in the same metric: 1) Andreas Krause at ETH (17.5), 2) Bernhard Schölkopf at Max Planck (14.9), 3) Peter Richtárik at KAUST (14), 4) Volkan Cevher at EPFL (12.5), 5) Max Welling at Amsterdam (12.2), 6) Stephan Günnemann at TU Munich (12.1), 7) Yishay Mansour at Tel Aviv (9.6), 8) Shie Mannor at Technion (9.3), 9) Francis Bach at ENS (9.2), 10) Shimon Whiteson at Oxford (8.9), 11) José Miguel Hernández-Lobato at Cambridge (8.8).

⁷<https://papercopilot.com/paper-list/neurips-paper-list/neurips-2022-paper-list/> (my team had 12 papers accepted; I was a coauthor on 9)

⁸<https://research.com/u/peter-richtarik>

⁹For the best paper published in Computational Optimization and Applications in 2020.

¹⁰<https://sineus.siam.org/Details-Page/10-most-highly-cited-articles-from-simax-since-2019-1>

¹¹From Robin.ly LinkedIn Post: “We are interviewing the world’s leading AI academics this week at NeurIPS2019. Look forward to sharing much more on the state of AI research, how it’s fueling AI commercialization & what we can expect from AI

2019	Best NeurIPS Reviewer Award ¹²
2019	Distinguished Speaker Award , Int. Conf. on Continuous Optimization, Berlin, Germany
2018	Best NeurIPS Reviewer Award ¹³
2018	2nd Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]
2018	6th Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [44]
2017	1st Most Read Paper in “Optimization Methods and Software” for paper [41]
2017	1st Most Downloaded Paper in “SIAM J. on Matrix Analysis and Applications” for paper [39]
2017	1st Most Trending Paper in “Mathematical Programming” for paper [10]
2017	Announcement of “Federated Learning” by Google (based on papers [51, 52])
2016–2017	2nd Most Downloaded Paper in “SIAM J. on Optimization” for paper [21]
2016	SIAM SIGEST Outstanding Paper Award for paper [21]
2016	EUSA Best Research or Dissertation Supervisor Award ¹⁴ (2nd Prize)
2016–now	Turing Fellow, The Alan Turing Institute , London
2016	EPSRC Fellowship in Mathematical Sciences ¹⁵
2014	Nominated for the Chancellor’s Rising Star Award ¹⁶ , University of Edinburgh
2013	Simons Institute Visiting Scientist Fellowship , UC Berkeley
2013	Nominated for the 2014 Microsoft Research Faculty Fellowship ¹⁷
2011 & 2012	Nominated for the Innovative Teaching Award , University of Edinburgh
2011–2017	Honorary Fellow , Heriot-Watt University
2007	CORE Fellowship , Université catholique de Louvain
2002	Cornell University Graduate Fellowship
2001	Dean’s Prize and Rector’s Prize , Comenius University
1992–2001	Winner of Numerous Mathematical Olympiads and Competitions

7. GRANTS

7.1 MY GRANTS¹⁸

2026–2028	\$410,413 (PI) , Industrial Research Grant from Saudi company Tahakom, “Optimizing LLM Efficiency through Precision and Parameter Reduction”, Joint with: Marco Canini (KAUST, PI) and Suhaib Fahmy (KAUST, PI)
2026–2027	\$100,000 (PI) , KAUST Small Translational Grant (sTRG), “TRACE: Transforming Textbooks into Verified, Interactive Courses using Retrieval-Augmented Generation”, Joint with: Laurent Condat (research scientist), Ivan Ilin (PhD student)
2026–2027	\$650,000 (PI) , KAUST Baseline Research Grant ¹⁹
2025–2027	\$100,000 , Google PhD Fellowship in Algorithms & Optimization for Kaja Gruntkowska ²⁰
2025–2028	\$675,000 (PI) , Competitive Research Grant (CRG)
2025–2028	1,600,000 SAR (PI) , RDIA Baseload Applied Grant (BAG)
2025–2026	\$650,000 (PI) , KAUST Baseline Research Grant

in the next decade. Spotlight interviews with Yoshua Bengio, Peter Richtárik, Charles Onu, Max Welling, Shimon Whiteson, Sharon Zhou, Liwei Wang, Song Han & many more. ”

¹²“Thank you for all your hard work reviewing for NeurIPS 2019! We are delighted to inform you that you were one of the 400 highest-scoring reviewers this year! You will therefore be given access (for a limited period of time) to one free registration to this year’s conference; you will later receive additional information by email explaining how to access your registration. ”

¹³“We are delighted to inform you that you were one of the 218 highest-scoring reviewers this year! You will therefore be given access (for a limited period of time) to one free registration to this year’s conference.”

¹⁴EUSA = Edinburgh University Students’ Association. One first and one second prize are given each year across all disciplines and levels of seniority at the University of Edinburgh.

¹⁵In total, 5 fellowships in mathematics were awarded in the UK in this round at all levels of seniority.

¹⁶One of two nominated from the School of Mathematics.

¹⁷Selected universities can nominate a single candidate. No European scientists got the award in 2014.

¹⁸All small grants (value below \$10k) are excluded from this list. The total value of the 16 small grants excluded is £42,090. Funding from the VCC and ECRC centers at KAUST is excluded from this list.

¹⁹Unrestricted basic research funding offered each year to KAUST Professors.

²⁰An unrestricted gift of \$50,000 per year for two years

2024–2029	\$11,000,000 (Co-PI) , KAUST Center of Excellence for Generative AI
2024–2025	\$650,000 (PI) , KAUST Baseline Research Grant
2024	1,000,000 SAR (PI) , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence
2023–2024	\$60,000 (PI) , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence
2023–2024	\$650,000 (PI) , KAUST Baseline Research Grant
2022–2023	\$60,000 (PI) , SDAIA-KAUST Center of Excellence in Data Science and Artificial Intelligence
2022–2023	\$40,000 (PI) , Top-up to KAUST Baseline Research Grant
2022–2023	\$540,000 (PI) , KAUST Baseline Research Grant
2021–2022	\$540,000 (PI) , KAUST Baseline Research Grant
2021–2022	\$100,000 (PI) , AI Initiative Funding
2020–2021	\$540,000 (PI) , KAUST Baseline Research Grant
2020	\$100,000 (PI) , AI Initiative Seed Funding, “Algorithmic, Systems and Privacy Aspects of Split Learning”, Joint with: Marco Canini (KAUST, Co-I) and Panos Kalnis (KAUST, Co-I)
2019–2020	\$200,000 (PI) , Extreme Computing Research Center (ECRC) funding, KAUST, “Optimization for Machine Learning”, Joint with: Tong Zhang (HKUST, PI)
2019–2020	\$540,000 (PI) , KAUST Baseline Research Grant
2018–2019	£216,843 (Co-I) , Innovate UK Grant, “Renewable Energy Performance Score (REPScore)”, Joint with: Enian (PI), Daniel Friedrich (Edinburgh, PI)
2018–2021	\$974,789 (Co-I) , CRG2017 Grant, “Analyzing Large Scale 3D Shape Collections”, Joint with: Peter Wonka (KAUST, PI), Maks Ovsjanikov (École Polytechnique, Co-I)
2017–2019	RUB 7,960,000 (PI) , Visiting Professor Grant, Moscow Institute of Physics and Technology
2018	\$10,000 (PI) , KICP grant in support of KAUST Research Workshop on Optimization and Big Data, 2018
2018–2019	\$400,000 (PI) , KAUST Baseline Research Grant ²¹
2017–2018	\$79,281 (PI) , KAUST Office of Sponsored Research Conference Support Grant URF/1/3347-01, “Optimization and Big Data”, Joint with: Marco Canini (KAUST, PI)
2016–2020	£70,000 EPSRC CASE²² PhD Studentship for Filip Hanzely
2017–2018	\$400,000 (PI) , KAUST Baseline Research Grant
2016–2017	\$133,333 (PI) , KAUST Baseline Research Grant (4 months of cover: March-June 2017)
2016–2020	£45,000 (PI) , Amazon Research Grant
2016–2020	£823,211 (PI) , EPSRC Early Career Fellowship in Mathematical Sciences EP/N005538/1, “Randomized Algorithms for Extreme Convex Optimization”
2016–2020	\$20,000 (PI) , Amazon EC2 Grant (partner funding associated with the EPSRC Fellowship)
2015	£20,000 (PI) , Alan Turing Institute Scoping Workshop Grant, “Distributed Machine Learning and Optimization”, Joint with: Artur Czuma (Warwick, PI), Ilias Diakonikolas (Edinburgh, PI), Mark Girolami (Warwick, PI), Raphael Hauser (Oxford, PI), John Shawe-Taylor (UCL, PI)
2015	£12,000 (PI) , Alan Turing Institute Scoping Workshop Grant, “Theoretical and Computational Approaches to Large Scale Inverse Problems”, Joint with: Simon Arridge (UCL, PI), John Aston (Cambridge, PI), Carola-Bibiane Schönlieb (Cambridge, PI), Andrew Stuart (Warwick, PI), Jared Tanner (Oxford, PI)
2014–2017	\$180,000 , Google Europe Doctoral Fellowship for Jakub Konečný
2013–2015	£125,849 (PI) , EPSRC First Grant EP/K02325X/1, “Accelerated Coordinate Descent Methods for Big Data Optimization”
2014–2015	£40,000 (PI) , School of Mathematics Grant, “Accelerated Coordinate Descent Methods for Big Data Optimization”, matching funding for my postdoc Z. Qu
2013	£18,785 (PI) , NAIS Travel Grant, my 2 students spending semester at Berkeley
2012–2014	£66,300 (PI) , NAIS Lecturer Grant, paying for a proportion of my time
2012–2014	£10,000 (PI) , NAIS Startup Grant

²¹Unrestricted basic research funding offered each year to KAUST Associate Professors.

²²CASE = Cooperative Awards in Science and Engineering

2012–2013	£49,518 (Co-I) , EPSRC grant EP/J020567/1, “Algorithms for Data Simplicity”, Joint with: Jared Tanner (Oxford, PI)
2011–2014	£646,264 (Co-I) , EPSRC and RCUK grant EP/I017127/1, “Mathematics for Vast Digital Resources”, Joint with: Burak Büke (Edinburgh, Co-I) and Jacek Gondzio (Edinburgh, PI)

7.2 GRANTS I HELPED TO PREPARE²³

2014–2019	£42 million + £5 million, “The Alan Turing Institute”. I am one of a small number of people who helped to prepare Edinburgh’s bid.
2014–2023	£4.5 million, EPSRC grant, “Maxwell Institute Graduate School in Mathematical Analysis and Applications”, PI: Anthony Carbery (Edinburgh). I am one of the named PhD supervisors on the grant.
2014–2021	£5.03 million, EPSRC grant, “Centre for Doctoral Training in Data Science”, PI: Chris Williams (Edinburgh). I am one of 45 named potential PhD advisors at U of Edinburgh.

8. MY TEAM

8.1 MY TEAM @ KAUST

08/2026–now	MS student: Tigran Melkonyan (from AUA, Armenia)
07/2026–now	Research scientist: Xun Qian (from Shanghai, China)
07/2026–now	Postdoc: Egor Shulgin (continuing after PhD at KAUST)
06/2026–now	Intern: Jan Niklas Kolzenburg (from Frankfurt, Germany)
06/2026–now	Postdoc: Grigory Malinovsky (from MIPT, Russia)
12/2025–05/2026	Postdoc: Artavazd Maranjyan (from KAUST, Saudi Arabia) ²⁴
08/2025–now	MS student: Alina Abdikarimova (from Nazarbayev U, Kazakhstan)
08/2025–now	MS student: Alexander Gaponov (from HSE, Russia)
08/2025–now	MS student: Zhirayr Tovmasyan (from Yerevan State U, Armenia)
08/2025–now	PhD student: Ammar Mahran (from U of Oxford, UK)
06/2025–now	PhD student: Yassine Maziane (from University of Liege, Belgium)
05/2025–08/2025	Intern: Alexander Gaponov (from HSE, Russia)
07/2025–05/2026	Intern: Igor Klimczak (from U of Oxford, UK)
05/2025–09/2025	Intern: Hussein Rammal (from École Polytechnique, France)
04/2025–09/2025	Intern: Adrien Fradin (from École Polytechnique, France)
01/2025–now	PhD student: Ivan Ilin (from Novosibirsk State University, Russia)
10/2024–04/2025	Intern: Philip Zmushko (from MIPT, Russia)
09/2024–07/2025	Intern: Zhirayr Tovmasyan (from Yerevan State University, Armenia)
06/2024–11/2024	Intern: Aadi Rane (from UC Berkeley, USA)
05/2024–09/2024	Intern: Wojciech Anyszka (from U of Groningen, Netherlands)
02/2024–05/2026	Postdoc: Sarit Khirirat (from KTH, Sweden)
02/2024–05/2024	Intern: Simone Maria Giancola (from Bocconi, Italy)
01/2024–05/2024	Intern: Dymitr Lubczyk (from Amsterdam, Netherlands)
01/2024–04/2024	Intern: Kirill Acharya (from MIPT, Russia)
01/2024–05/2024	Intern: Robin Yadav (from UBC, Canada)
01/2024–now	PhD student: Kaja Gruntkowska (from U of Oxford, UK)
12/2023–03/2024	Intern: Anh-Duc Nguyen (from NUS, Singapore)
10/2023–03/2024	Intern: Timur Kharisov (from MIPT, Russia)
10/2023–04/2024	Intern: Georg Meinhardt (from University of Oxford, UK)
09/2023–01/2024	Intern: Marta Pozzi (from Pavia, Italy)

²³Large grants which I helped to prepare but where I am not formally an investigator.

²⁴next position: Postdoc, EPFL, Switzerland

09/2023–02/2024	Intern: Ákos Zahorský (from Eötvös Loránd University, Hungary)
08/2023–05/2025	MS student: Omar Shaikh Omar (from University of Washington, USA)
08/2023–12/2024	MS student: Artem Riabinin (from Lomonosov Moscow State U, Russia)
08/2023–12/2025	PhD student: Artavazd Maranjyan (from Yerevan State U, Armenia)
07/2023–09/2023	Intern: Andrei Panferov (from MIPT, Russia)
06/2023–08/2023	Intern: Babis Kostopoulos (from U of Athens, Greece)
04/2023–09/2023	Intern: Ahmad Rammal (from École Polytechnique, France)
01/2023–05/2023	Intern: Dinis Seward (from University of Oxford, UK)
01/2023–12/2024	MS student: Ivan Ilin (from Novosibirsk State University, Russia)
01/2023–now	PhD student: Hanmin Li (from USTC, China)
09/2022–now	Postdoc: Yury Demidovich (from MIPT, Russia)
09/2022–now	PhD student: Abdurakhmon Sadiev (from MIPT, Russia)
08/2022–10/2023 ²⁵	MS student: Rafał Szlendak (from Warwick University, UK)
08/2022–02/2023	Intern: Wenzhi “Tom” Fang (from ShanghaiTech University, China)
07/2022–08/2022	Intern: Omar Shaikh Omar (from University of Washington, USA)
07/2022–10/2022	Intern: Michał Grudzień (from Oxford, UK)
06/2022–01/2023	Intern: Artavazd Maranjyan ²⁶ (from Yerevan State University, Armenia)
06/2022–09/2022	Intern: Kaja Gruntkowska (from Warwick, UK)
06/2022–now	PhD student: Igor Sokolov (continuing after MS at KAUST)
01/2022–07/2022	Intern: Abdurakhmon Sadiev (from MIPT, Russia)
01/2022–04/2025	PhD student: Kai Yi (from Xi’an Jiaotong University, China)
01/2022–06/2026	PhD student: Grigory Malinovsky (from MIPT, Russia) ²⁷
11/2021–06/2024	Postdoc: Avetik Karagulyan (from CREST, France)
11/2021–02/2022	Intern: Navish Kumar (from IIT Kharagpur, India)
09/2021–07/2026	PhD student: Egor Shulgin (continuing after MS at KAUST)
07/2021–11/2021	Intern: Muhammad Harun Khan (from Imperial College, UK)
07/2021–10/2021	Intern: Rafał Szlendak (from Warwick University, UK)
06/2021–06/2024	Postdoc: Alexander Tyurin (from MIPT, Russia)
06/2021–08/2021	Intern: Bokun Wang (from UC Davis, USA)
03/2021–06/2024	PhD student: Lukang Sun (from Nanjing University, China)
03/2021–08/2021	Intern: Rustem Islamov ²⁸ (from TU Munich, Germany)
03/2021–11/2021	Intern: Ilyas Fatkhullin ²⁹ (from TU Munich, Germany)
01/2021–11/2023	PhD student: Slavomír Hanzely (continuing after MS at KAUST)
09/2020–03/2022	Research scientist: Zhize Li (from Tsinghua University, China)
10/2020–03/2021	Intern: Bokun Wang (from UC Davis, USA)
09/2020–02/2021	Intern: Eduard Gorbunov (from MIPT, Russia)
08/2020–05/2025	PhD student: Konstantin Burlachenko (from Bauman Moscow State Tech U, Russia)
08/2020–05/2022	MS student: Igor Sokolov (from MIPT, Russia)
08/2020–12/2021	MS student: Grigory Malinovsky (from MIPT, Russia)
08/2020–09/2020	Intern: Wenlin Chen (from University of Manchester, UK)
06/2020–11/2020	Intern: Rustem Islamov (from MIPT, Russia)
05/2020–06/2020	Intern: Othmane Sebbouh (from École Polytechnique, France)
05/2020–10/2020	Intern: Ahmed Khaled Ragab (from Cairo University, Egypt)
02/2020–12/2020	Research scientist: El Houcine Bergou (from Toulouse, France)
02/2020–08/2021	MS student: Egor Shulgin (from MIPT, Russia)
02/2020–03/2020	Intern: Eduard Gorbunov (from MIPT, Russia)
01/2020–02/2020	Intern: Alexander Rogozin (from MIPT, Russia)

²⁵Dropped out of MS studies to join a Large Language Model startup in Germany.

²⁶I am supervising Arto’s MS thesis at Yerevan State University.

²⁷next position: Postdoc, KAUST; then Postdoc, EPFL, from Nov 2026

²⁸I have supervised Rustem Islamov’s BS thesis at MIPT. Rustem is now an MS student at Institut Polytechnique de Paris, France.

²⁹I have supervised Ilyas Fatkhullin’s MS thesis at TU Munich. Ilyas is now a PhD student at ETH Zürich Switzerland.

01/2020–02/2020	Intern: Aleksandr Beznosikov (from MIPT, Russia)
01/2020–02/2020	Intern: Grigory Malinovsky (from MIPT, Russia)
01/2020–05/2025	PhD student: Elnur Gasanov (continuing after MS from KAUST)
01/2020–09/2022	PhD student: Dmitry Kovalev (continuing after MS from KAUST)
11/2019–now	Research scientist: Laurent Condat (from Grenoble, France)
10/2019–11/2022	Postdoc: Mher Safaryan (from Yerevan State University, Armenia)
09/2019–08/2020	Postdoc: Zhize Li (from Tsinghua University, China)
08/2019–12/2020	MS student: Alyazeed Basyoni (from Carnegie Mellon University, USA)
08/2019–12/2020	MS student: Slavomír Hanzely (from Comenius University, Slovakia)
06/2019–09/2019	Intern: Ahmed Khaled Ragab (from Cairo University, Egypt)
03/2019–09/2019	Intern: Sélim Chraïbi (from Grenoble, France)
02/2019–10/2021	Postdoc: Adil Salim (from Télécom ParisTech, France)
02/2019–03/2019	Intern: Ľudovít Horváth (from Comenius University, Slovakia)
01/2019–02/2019	Intern: Dmitry Kamzolov (from MIPT, Russia)
01/2019–02/2019	Intern: Vladislav Elsukov (from MIPT, Russia)
01/2019–02/2019	Intern: Igor Sokolov (from MIPT, Russia)
01/2019–02/2019	Intern: Egor Shulgin (from MIPT, Russia)
01/2019–02/2019	Intern: Eduard Gorbunov (from MIPT, Russia)
01/2019–03/2022	PhD student: Alibek Sailanbayev (continuing after MS from KAUST) ³⁰
01/2019–06/2022	PhD student: Samuel Horváth (continuing after MS from KAUST)
11/2018–11/2021	Postdoc: Xun Qian (from Hong Kong Baptist University, Hong Kong)
09/2018–12/2019	MS student: Elnur Gasanov (from MIPT, Russia)
09/2018–12/2019	MS student: Dmitry Kovalev (from MIPT, Russia)
03/2018–08/2018	Intern: Sarah Sachs ³¹ (from TU Munich, Germany)
01/2018–02/2018	Intern: Eduard Gorbunov (from MIPT, Russia)
01/2018–02/2018	Intern: Elnur Gasanov (from MIPT, Russia)
01/2018–02/2018	Intern: Dmitry Kovalev ³² (from MIPT, Russia)
01/2018–02/2018	Intern: Slavomír Hanzely ³³ (from Comenius University, Slovakia)
01/2018–01/2019	Postdoc: El Houcine Bergou (from Institut National Polytechnique, Toulouse, France)
10/2017–11/2017	Intern: Nikita Doikov (from HSE Moscow, Russia)
08/2017–12/2017	PhD student: Viktor Lukáček ³⁴ (from Comenius University, Slovakia)
08/2017–12/2021	PhD student: Konstantin Mishchenko (from ENS, France)
08/2017–11/2020	PhD student: Filip Hanzely (now: Quant, Wincent)
08/2017–12/2018	MS student: Alibek Sailanbayev (from Nazarbayev University, Kazakhstan)
08/2017–12/2018	MS student: Samuel Horváth (from Comenius University, Slovakia)
05/2017–05/2019	Postdoc: Aritra Dutta (from University of Central Florida, USA)
05/2017–07/2017	Intern: Atal Sahu (from IIT Kanpur, India)
05/2017–07/2017	Intern: Aashutosh Tiwari (from IIT Kanpur, India)

8.2 MY TEAM @ Kempelen Institute for Intelligent Technologies (KInIT)

09/2022–10/2023	PhD student: Ivan Agarský (from Comenius University, Slovakia)
-----------------	--

8.3 MY TEAM @ Mohammed bin Zayed University of Artificial Intelligence (MBZUAI)

09/2022–08/2023	Postdoc: Sarit Khirirat (from KTH, Sweden)
-----------------	--

³⁰Was forced to drop out of PhD due to serious personal/family reasons.

³¹I have supervised Sarah Sachs' MS thesis at TU Munich.

³²I have supervised Dmitry Kovalev's BS thesis at MIPT.

³³I have supervised Slavomír Hanzely's BS thesis at Comenius University.

³⁴Viktor Lukáček left after spending 1 semester at KAUST as he realized PhD was not the right path for him.

8.4 MY TEAM @ MOSCOW INSTITUTE OF PHYSICS AND TECHNOLOGY

09/2020–12/2021	PhD student: Eduard Gorbunov (PhD at MIPT, co-advised with Alexander Gasnikov; thesis: Distributed and Stochastic Optimization Methods with Gradient Compression and Local Steps , 2021)
09/2018–10/2019	Dmitry Kamzolov
09/2018–10/2019	Vladislav Elsukov
09/2018–10/2019	Igor Sokolov (now: PhD student in my team at KAUST)
08/2018–10/2019	Egor Shulgin (now: postdoc in my team at KAUST)
10/2017–10/2019	Eduard Gorbunov (now: Assistant Professor of Statistics and Data Science at MBZUAI since 08/2025; previously Postdoc, 09/2022–03/2024, and Research Scientist, 04/2024–07/2025, at MBZUAI)
10/2017–08/2018	Dmitry Kovalev (now: Postdoc at Université catholique de Louvain)
10/2017–08/2018	Elnur Gasanov (now: PhD student in my team at KAUST)

8.5 MY TEAM @ UNIVERSITY OF EDINBURGH

09/2016–07/2017	PhD student: Filip Hanzely (transferred to KAUST after 1 year in Edinburgh to follow me, with an MS degree with distinction)
03/2016–07/2016	Postdoc: Robert M. Gower
10/2015–06/2019	PhD student: Nicolas Loizou (now: Postdoc, MILA, Montréal)
10/2015–02/2017	PhD student: Theo Pavlakou (now: Google; 2nd advisor; main advisor: Iain Murray)
03/2015–03/2016	PhD student: Robert M. Gower (now: Assistant Prof. at Télécom ParisTech)
03/2015–06/2015	Visiting PhD student: Luca Bravi (from University of Florence)
10/2014–03/2015	Postdoc: Ademir Ribeiro (now: Associate Prof. at University of Paraná)
09/2014–11/2017	PhD student: Dominik Csiba (now: Algo Lead at Nozdormu, Slovakia)
08/2013–07/2017	PhD student: Jakub Konečný (now: Research Scientist, Google)
12/2013–08/2015	Postdoc: Zheng Qu (now: Assistant Prof. at University of Hong Kong)
09/2012–02/2013	Visiting PhD student: Minnan Luo (now: Associate Prof. at Xi'an Jiaotong University)
10/2012–07/2014	Postdoc: Olivier Fercoq (now: Assistant Prof. at Télécom ParisTech)
02/2012–07/2014	Postdoc: Rachael Tappenden (now: Assistant Prof. at University of Canterbury)
01/2012–06/2012	Postdoc: Jakub Mareček (now: IBM Research, Dublin)
09/2010–03/2014	PhD student: Martin Takáč (now: Associate Prof. at Mohammed bin Zayed University of Artificial Intelligence, UAE)
2010–2015	Supervised 20 MSc Dissertations
2010–2015	Supervised 7 undergraduate students supported by research scholarships (EPSRC, Nuffield, College, ...)

8.6 MY TEAM: AWARDS, RECOGNITIONS & NOTABLE ACHIEVEMENTS³⁵

2026 (Mahran)	Dean's List ³⁶ , Applied Mathematics and Computational Science, KAUST
2026 (Malinovsky)	Dean's List, Applied Mathematics and Computational Science, KAUST
2026 (Shulgin)	Dean's List, Applied Mathematics and Computational Science, KAUST
2026 (Gruntkowska)	Dean's List, Statistics, KAUST
2026 (Sadiev)	Dean's List, Computer Science, KAUST
2026 (Li)	Dean's List, Computer Science, KAUST
2026 (Sokolov)	ICLR 2026 Notable Reviewer
2026 (Malinovsky)	Best Paper Award (runner-up), International Conference on Computational Optimization, Abu Dhabi, UAE (for paper [245])
2025 (Malinovsky)	NeurIPS 2025 Top Reviewer

³⁵All travel grant awards are excluded.

³⁶Carries a 2,500 USD cash prize.

2025 (Condat)	Senior Area Editor, IEEE Transactions on Signal Processing
2025 (Fradin)	Research Internship Prize, École Polytechnique
2025 (Sadiev)	2 papers accepted at NeurIPS 2025
2025 (Gruntkowska)	Google PhD Fellowship in Algorithms and Optimization³⁷ – first³⁸ in GCC
2025 (Malinovsky)	ICLR 2025 Notable Reviewer
2025 (Mahran)	Dean’s Award (Computer Science) ³⁹ , KAUST
2025 (Malinovsky)	Dean’s List ⁴⁰ , Applied Mathematics and Computational Science, KAUST
2025 (Sokolov)	Dean’s List, Applied Mathematics and Computational Science, KAUST
2025 (Gruntkowska)	Dean’s List, Statistics, KAUST
2025 (Burlachenko)	Dean’s List, Computer Science, KAUST
2025 (Maranjyan)	Dean’s List, Computer Science, KAUST
2025 (Shulgin)	Dean’s List, Computer Science, KAUST
2025 (Burlachenko)	Accepts a Quantitative Research Scientist position at Abu Dhabi Investment Authority (ADIA), Abu Dhabi, UAE
2025 (Gasarov)	Accepts a Research Scientist position at Saudi Aramco, KSA
2025 (Yi)	Accepts a Research Scientist position at META, California, USA
2025 (Condat)	World’s Top 2% Scientist by Stanford
2024 (Tyurin)	Joins Skoltech as an Assistant Professor & AI Research Institute (AIRI) as a Team Leader
2024 (Yi)	Oral at NeurIPS 2024
2024 (Ilin)	Oral at NeurIPS 2024
2024 (Burlachenko)	Oral at NeurIPS 2024
2024 (Tyurin)	4 papers accepted at NeurIPS 2024
2024 (Tyurin)	Spotlight at NeurIPS 2024
2024 (Gruntkowska)	Spotlight at NeurIPS 2024
2024 (Gruntkowska)	Nominated by KAUST for Apple Doctoral Scholarship
2024 (Malinovsky)	ICML 2024 Best Reviewer
2024 (Gruntkowska)	Dean’s List ⁴¹ , Statistics, KAUST
2024 (Malinovsky)	Dean’s List, Applied Mathematics and Computational Science, KAUST
2024 (Sadiev)	Dean’s List, Computer Science, KAUST
2024 (Condat)	World’s Top 2% Scientist by Stanford
2024 (Malinovsky)	NeurIPS 2024 Top Reviewer
2024 (Malinovsky)	Summer of Reproducibility 2023 Flower AI Award ⁴² for implementing the TAMUNA algorithm (based on paper [217])
2024 (Agarský)	Summer of Reproducibility 2023 Flower AI Award for implementing the TAMUNA algorithm (based on paper [217])
2023 (Malinovsky)	Nominated by KAUST for Apple PhD Scholarship ⁴³
2023 (Tyurin)	4 papers accepted at NeurIPS 2023
2023 (Sun)	Dean’s List ⁴⁴ , Computer Science, KAUST
2023 (Burlachenko)	Dean’s List, Computer Science, KAUST
2023 (Sadiev)	Dean’s List, Computer Science, KAUST
2023 (S. Hanzely)	Dean’s List, Applied Mathematics and Computational Science, KAUST
2023 (Malinovsky)	Dean’s List, Applied Mathematics and Computational Science, KAUST
2023 (Mishchenko ⁴⁵)	ICML 2023 Outstanding Paper Award (0.09% success rate)

³⁷100,000 USD unrestricted gift to fund Kaja’s PhD for 2 years.

³⁸Three more were awarded to students in the UAE in 2025.

³⁹A financial add-on to the KAUST Fellowship, worth 6,000 USD annually, given to a few best incoming students by the Dean.

⁴⁰Carries a 2,500 USD cash prize.

⁴¹In 2024, given to top 20% students at KAUST annually. Carries a 2,500 USD cash prize.

⁴²\$ 2,000 Cash Prize shared with Ivan Agarský.

⁴³Several invited institutions can nominate up to 3 students.

⁴⁴Given to top 25% students at KAUST annually. Carries a 2,500 USD cash prize.

⁴⁵Former PhD student

2023 (Mishchenko)	Action Editor, Transactions on Machine Learning Research (TMLR)
2023 (Gruntkowska)	Dean's Award (Statistics) ⁴⁶ , KAUST
2023 (Maranjyan)	Dean's Award (Computer Science), KAUST
2023 (Mishchenko)	Joins Samsung AI, Cambridge, UK, as a Research Scientist
2023 (Condat)	World's Top 2% Scientist by Stanford
2023 (Fatkhullin ⁴⁷)	NeurIPS 2023 Top Reviewer
2022 (Kovalev)	6 papers accepted at NeurIPS 2022
2022 (Burlachenko)	Grant from AMD Inc (two GPUs)
2022 (Beznosikov ⁴⁸)	NeurIPS 2022 Top Reviewer⁴⁹
2022 (Tyurin)	NeurIPS 2022 Top Reviewer
2022 (Gorbunov)	NeurIPS 2022 Top Reviewer
2022 (Malinovsky)	NeurIPS 2022 Top Reviewer
2022 (Sokolov)	NeurIPS 2022 Top Reviewer
2022 (Safaryan)	NeurIPS 2022 Top Reviewer
2022 (Mútny ⁵⁰)	NeurIPS 2022 Top Reviewer
2022 (Gower ⁵¹)	NeurIPS 2022 Top Reviewer
2022 (Kovalev)	Joins Université catholique de Louvain as a postdoc with Yurii Nesterov
2022 (Malinovsky)	2022 CEMSE Academic Excellence Award⁵²
2022 (Shulgin)	ICML 2022 Outstanding (Top 10%) Reviewer⁵³
2022 (Gasarov)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Gorbunov)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Khaled)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Condat)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Sadiev)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Tyurin)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Sokolov)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Horváth)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Szlendak)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Mishchenko)	ICML 2022 Outstanding (Top 10%) Reviewer
2022 (Sadiev)	Dean's Award, KAUST
2022 (Sadiev)	KAUST Doctoral Development Fellowship
2022 (Szlendak)	Dean's Award (Applied Mathematics), KAUST
2022 (Szlendak)	KAUST Doctoral Development Fellowship
2022 (Shulgin)	Research Internship at Apple, Cambridge, UK
2022 (S. Hanzely)	Research Internship at the Flatiron Institute, New York, USA
2022 (S. Hanzely)	Research Internship at the MBZUAI, Abu Dhabi, KSA
2022 (Malinovsky)	Research Internship at the CISPA Helmholtz Center for Information Security, Saarbrücken, Germany
2022 (Gorbunov)	Joins Mohamed bin Zayed University of Artificial Intelligence , Abu Dhabi, UAE, as a Postdoc
2022 (Horváth)	Joins Mohamed bin Zayed University of Artificial Intelligence , Abu Dhabi, UAE, as an Assistant Professor
2022 (Safaryan)	AISTATS 2022 Top Reviewer

⁴⁶A financial add-on to the KAUST Fellowship, worth 6,000 USD annually, given to a few best incoming students by the Dean.

⁴⁷Former intern.

⁴⁸Former intern.

⁴⁹<https://neurips.cc/Conferences/2022/ProgramCommittee>

⁵⁰Former intern.

⁵¹Former PhD student.

⁵²Carries a \$2,500 cash prize.

⁵³<https://icml.cc/Conferences/2022/Reviewers>

2022 (Loizou)	Joins The Johns Hopkins University as an Assistant Professor in the Department of Applied Mathematics & Statistics, and the Mathematical Institute for Data Science (MINDS), with a secondary appointment in Computer Science
2022 (Khaled)	Joins Princeton University as a PhD Student in the ECE Department ⁵⁴
2022 (Gower ⁵⁵)	Action Editor, Transactions of Machine Learning Research (TMLR)
2022 (Condat)	World's Top 2% Scientist by Stanford ⁵⁶
2021 (Malinovsky)	2021 CEMSE Student Research Excellence Award ⁵⁷
2021 (Kovalev)	2021 CEMSE Student Research Excellence Award ⁵⁸
2021 (Horváth)	2021 Al-Kindi Statistics Research Student Award ⁵⁹
2021 (Mishchenko)	Rising Stars in Data Science ; invited talk to a workshop at the University of Chicago ⁶⁰
2021 (Gorbunov)	NeurIPS 2021 Outstanding (Top 8%) Reviewer Award
2021 (Mishchenko)	NeurIPS 2021 Outstanding (Top 8%) Reviewer Award
2021 (Shulgin)	Research Internship at Samsung AI Research Center, Cambridge, UK
2021 (Horváth)	Research Internship at Facebook AI Research, Canada
2021 (Mishchenko)	ICML 2021 Top 10% Reviewer
2021 (Gorbunov)	ICML 2021 Top 10% Reviewer
2021 (Mishchenko)	ICML 2021 Expert Reviewer
2021 (F. Hanzely)	ICML 2021 Expert Reviewer
2021 (Gorbunov)	ICML 2021 Expert Reviewer
2021 (Condat)	ICML 2021 Expert Reviewer
2021 (Kovalev, Gasanov)	Best Student Paper Award at the International Workshop on Federated Learning for User Privacy and Data Confidentiality in Conjunction with ICML 2021 (for joint paper [166])
2021 (Kovalev)	Ilya Segalovich Prize for Young Researchers ⁶¹
2021 (Mishchenko)	Most Popular Spotlight Talk (2nd place) at KAUST Conference on AI
2021 (Mishchenko)	ICLR 2021 Outstanding Reviewer Award
2021 (Gorbunov)	ICLR 2021 Outstanding Reviewer Award
2021 (Condat)	Associate Editor, IEEE Transactions on Signal Processing
2021 (Condat)	World's Top 2% Scientist by Stanford
2020 (F. Hanzely)	Joined Toyota Tech. Institute at Chicago as a Research Assistant Professor
2020 (Mishchenko)	2020 CEMSE Student Research Excellence Award ⁶²
2020 (Horváth)	Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning (for joint paper [135])

⁵⁴<https://rka97.github.io>

⁵⁵Former PhD student.

⁵⁶https://elsevier.digitalcommonsdata.com/datasets/btchxktzyw/4?fbclid=IwAR0u4xhKMKGIsi_prZLx0IOaMPzV-LNCmoILDYua90eybIViYtE6S170vyc

⁵⁷Given annually to a handful of the best Applied Mathematics students at KAUST. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science." Carries a \$1,000 cash prize.

⁵⁸Given annually to a handful of the best Computer Science students. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science."

⁵⁹Given annually to a handful of the best Statistics students at KAUST. "The recipients exemplify the highest of academic standards and represent our confidence in your future contributions to the KAUST community, academia, and science."

⁶⁰The Rising Stars in Data Science workshop at the University of Chicago focuses on celebrating and fast tracking the careers of exceptional data scientists at a critical inflection point in their career: the transition to postdoctoral scholar, research scientist, industry research position, or tenure track position. An event associated with the The Center for Data and Computing (CDAC) at the University of Chicago.

⁶¹Four awards were given; each award carries a cash prize of 1,000,000 RUB ($\approx$ 14,000 USD).

⁶²One of 5 awards given to KAUST Computer Science students.

2020 (Loizou)	Runner Up for OR Society Best Doctoral Dissertation Prize ⁶³ (for year 2019)
2020 (Horváth)	NeurIPS 2020 Best Reviewer Award ⁶⁴
2020 (Gorbunov)	NeurIPS 2020 Best Reviewer Award
2020 (F. Hanzely)	NeurIPS 2020 Best Reviewer Award
2020 (Condat)	NeurIPS 2020 Best Reviewer Award
2020 (Khaled)	NeurIPS 2020 Best Reviewer Award
2020 (Horváth)	Research Internship at Samsung AI Research Center, Cambridge, UK
2020 (Mishchenko)	Research Internship at Google, USA (performed remotely due to Covid-19)
2020 (Kovalev)	Ilya Segalovich Prize for Young Researchers ⁶⁵
2020 (Burlachenko)	Dean's Award, KAUST
2020 (Malinovsky)	Dean's Award, KAUST
2020 (Mishchenko)	AAAI 2020 Outstanding Program Committee Member Award (awarded to top 12 out of over 6,000 reviewers)
2020 (Condat)	World's Top 2% Scientist by Stanford
2019 (Mishchenko)	NeurIPS 2019 Best Reviewer Award
2019 (S. Hanzely)	Dean's Award, KAUST
2019 (F. Hanzely)	Research Internship at Google, New York
2019 (Horváth)	Research Internship at Amazon, Berlin
2019 (Sailanbayev)	Research Internship at Intel, Germany
2018 (Kovalev)	Dean's Award, KAUST
2018 (Loizou)	Research Internship at Facebook AI Research (FAIR), Montréal
2018 (Mishchenko)	Research Internship at Amazon, Seattle
2018 (F. Hanzely)	Research Internship at Microsoft Research (with Lin Xiao)
2018 (F. Hanzely)	Research Internship at Amazon, Berlin, Scalable Machine Learning Group
2018 (Horváth)	Best DS³ Poster Award ⁶⁶ , Paris (1st Prize; for joint paper [81])
2018 (Doikov)	Best Talk Award ⁶⁷ , Voronovo, Russia (1st Prize; for joint paper [69])
2018 (F. Hanzely)	WEP Best Poster Award (3rd Place), KAUST
2017 (Mishchenko)	Dean's Award, KAUST
2017 (Lukáček)	Dean's Award, KAUST
2017 (F. Hanzely)	Dean's Award, KAUST
2017 (Gower)	18th IMA Leslie Fox Prize ⁶⁸ (2nd Prize; for joint paper [39])
2016 (Csiba)	Postgraduate Essay Prize, School of Mathematics, University of Edinburgh
2016 (F. Hanzely)	CASE PhD Studentship (£93,333 award; 3/4 from EPSRC, 1/4 from Amazon)
2016 (Loizou)	A. G. Leventis Foundation Grant for PhD studies
2015 (Takáč)	OR Society Best Doctoral Dissertation Prize (for year 2014)
2015 (Loizou)	A. G. Leventis Foundation Grant for PhD studies
2015 (Loizou)	Principal's Career Development Scholarship ⁶⁹ (in Data Science)
2015 (Kisiala)	Best Student Prize ⁷⁰ , OR MSc Programme, School of Mathematics, Edinburgh

⁶³The OR Society was created in April 1948 as the Operational Research Club, becoming the OR Society in 1953. It is the world's oldest-established learned society catering to the OR profession and one of the largest in the world, with members in 53 countries https://en.wikipedia.org/wiki/Operational_Research_Society.

⁶⁴<https://icml.cc/Conferences/2020/Reviewers>

⁶⁵Nine awards were given in the area of Computer Science; each award carries a cash prize of 350,000 RUB ($\approx$ 5,000 USD).

⁶⁶DS³ stands for Data Science Summer School, held at École Polytechnique, Paris, during June 25–29, 2018. There were 170 posters in the competition, from MS and PhD students, and postdocs. Samuel's poster, based on joint work [81], won the main prize, which also attracted a 500 EUR check.

⁶⁷Event: "Traditional Youth School in Control, Information and Optimization", organized by Boris Polyak.

⁶⁸"The Leslie Fox Prize is a biennial prize established in 1985 by the IMA in honour of mathematician Leslie Fox (1918-1992). The prize honours young numerical analysts worldwide (any person less than 31 years old), and applicants submit papers for review. A committee [...] awards First Prize and Second Prizes based on mathematical and algorithmic brilliance in tandem with presentational skill"

⁶⁹Principal's Career Development Scholarship: A highly competitive scholarship offered to 3 incoming PhD students in mathematics at the University of Edinburgh each year.

⁷⁰For best performance in courses and MSc Dissertation, which I supervised.

2015 (Fercoq)	17th IMA Leslie Fox Prize (2nd Prize; for joint paper [21])
2015 (Csiba)	Best Contribution Award (2nd Prize; for joint paper [35]), Workshop: Optimization and Big Data, Edinburgh. Committee: Arkadi Nemirovskii (Georgia Tech) and Rodolphe Jenatton (Amazon)
2015 (Konečný)	BASP Frontiers Best Contribution Award (1st prize in the field of signal processing; for joint paper [20]), Villars-sur-Ollon, Switzerland
2014 (Konečný)	Google European Doctoral Fellowship ⁷¹ (\$180,000 unrestricted gift funding Jakub’s PhD for 3 years)
2014 (Csiba)	Principal’s Career Development Scholarship (in Data Science)
2013 (Konečný)	Principal’s Career Development Scholarship (in Data Science)
2013 (Takáč)	16th IMA Leslie Fox Prize (2nd Prize; for joint paper [10])
2013 (Takáč)	SIAM Certificate in Recognition of Outstanding Efforts and Accomplishments, on behalf of the SIAM Chapter at the University of Edinburgh for academic year 2012–2013
2012 (Takáč)	INFORMS Computing Society Best Student Paper Prize (sole runner up; for joint paper [8]), Phoenix, Arizona
2012 (Banks-Watson)	Best Student Prize, OR MSc Programme, School of Mathematics, Edinburgh
2012 (Takáč)	Best Talk Award , SIAM National Student Chapter Conference, Manchester, UK
2012 (Takáč)	Best Talk Award, Edinburgh Postgraduate Colloquium, University of Edinburgh
2012 (Takáč)	Alice Margaret Campbell Bequest Fund Award for success in 1st year of PhD
2011 (Takáč)	Certificate of Appreciation, 24th Biennial Conf. on Numerical Analysis, Glasgow, UK
2011 (Takáč)	Best Poster Award, SIAM Student Chapter Conference, Edinburgh, UK

8.7 MY TEAM: SELECTED PRIOR/INDEPENDENT ACHIEVEMENTS⁷²

2026 (Loizou)	NSF CAREER Award ⁷³
2026 (Gower)	Outstanding Paper Honorable Mention Award ⁷⁴ at ICLR 2026
2025 (Gorbunov)	Area Chair, NeurIPS
2024 (Gorbunov)	Action Editor, TMLR
2024 (Mishchenko)	Action Editor, TMLR
2024 (Takáč)	Senior Area Chair, NeurIPS
2023 (Gower)	Action Editor, TMLR
2023 (Záhorský)	Member, Slovak Committee of Mathematical Olympiad
2022 (Záhorský)	Coordinator, European Girls’ Mathematical Olympiad, Hungary
2022 (Condat)	Meritorious Service Award from the journal Mathematical Programming ⁷⁵
2022 (Burlachenko)	Second Place, KAUST Chess Tournament
2021 (Gaponov)	Bronze Medal, Belarusian National Olympiad in Competitive Programming
2021 (Maranjyan)	Outstanding Final Project Award ⁷⁶ , Yerevan State University, Armenia
2020 (Záhorský)	Deputy Leader, 14th Middle European Mathematical Olympiad, virtual

⁷¹Google quote: “Nurturing and maintaining strong relations with the academic community is a top priority at Google. Today, we’re announcing the 2014 Google PhD Fellowship recipients. These students, recognized for their incredible creativity, knowledge and skills, represent some of the most outstanding graduate researchers in computer science across the globe. We’re excited to support them, and we extend our warmest congratulations.”

⁷²These awards are independent of my input, and were in most cases obtained before joining my team.

⁷³Project: “Structural and Algorithmic Foundations of Variational Inequalities in Machine Learning: From Principles to Scalability”. https://www.nsf.gov/awardsearch/show-award/?AWD_ID=2542902

⁷⁴Two papers won the Main Prize, and one paper won the Honorable Mention Award. More than 19,000 papers were submitted to ICLR in 2026.

⁷⁵Mathematical Programming is the leading optimization journal. “The Meritorious Service Award was created to acknowledge these continued efforts. In 2022 our Editorial Board assessed the referees who have demonstrated exceptional diligence in their service to the journal.”

⁷⁶Awarded to 6 students from more than 250 students for best undergraduate thesis.

2020 (Basyoni)	National Deputy Leader and Head Coach at the International Olympiad of Informatics, Saudi Arabia
2019 (Záhorský)	Deputy Leader, 13th Middle European Mathematical Olympiad, Pardubice, Czech Republic
2019 (Panferov)	Gold Medal, International Physics Olympiad, Tel Aviv, Israel
2019 (Riabinin)	Winner, Phystech Olympiad in Physics, Dolgoprudny, Russia
2019 (Basyoni)	National Deputy Leader and Head Coach at the International Olympiad of Informatics, Saudi Arabia
2019 (Li)	Tsinghua Outstanding Doctoral Dissertation Award
2017 (Záhorský)	Silver Medal, 59th International Mathematical Olympiad, Cluj-Napoca, Romania (representing Slovakia)
2018 (Riabinin)	Prizewinner, Regional Stage of the All Russian Olympiad in Physics, Russia
2018 (Riabinin)	2nd Place, City Physics Olympiad, Nizhny Novgorod, Russia
2018 (Riabinin)	Prizewinner, Phystech Olympiad in Mathematics, Dolgoprudny, Russia
2018 (Mishchenko & Sailanbayev)	80th Place, 2018 IEEEExtreme programming competition ⁷⁷
2017 (Záhorský)	Honorable Mention, 58th International Mathematical Olympiad, Rio de Janeiro, Brazil (representing Slovakia)
2017 (Ilin)	Captain of the Russian team @ International Young Physicists Tournament, Singapore/Russia
2017 (Karagulyan)	Second Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2017 (S. Hanzely)	8–10th Place, Vojtech Jarník International Mathematical Competition (1st place among Czech and Slovak contestants)
2017 (Horváth)	37th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic
2016 (Záhorský)	Silver Medal, 10th Middle European Mathematical Olympiad, Vöcklabruck, Austria (representing Slovakia)
2016 (Karagulyan)	2nd Prize, Mirror of William Lowell Putnam Mathematical Competition
2016 (Malinovsky)	Abramov's Scholarship for students with the best grades at MIPT
2016 (S. Hanzely)	Participation, 57th International Mathematical Olympiad, Hong Kong
2016 (S. Hanzely)	3rd Place, Slovak National Mathematical Olympiad
2016 (S. Hanzely)	1st Place, Slovak Mathematical Olympiad, Regional Round
2016 (S. Hanzely)	1st Place, Slovak Informatics Olympiad, Regional Round
2016 (Horváth)	36th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic
2015 (Karagulyan)	Third Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2016 (Horváth)	3rd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2016 (Sailanbayev)	Semifinalist, ACM ICPC Programming Contest, NEERC region, Almaty, Kazakhstan
2015 (Karagulyan)	Second Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2015 (Karagulyan)	Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia
2015 (S. Hanzely)	Bronze Medal, Middle European Mathematical Olympiad
2015 (S. Hanzely)	2nd Place, Slovak Informatics Olympiad, Regional Round
2015 (Sailanbayev)	2nd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2015 (Mishchenko)	1st Prize, HSE Olympiad in Applied Mathematics and Informatics, Moscow, Russia
2014 (Karagulyan)	Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia
2014 (Malinovsky)	Bronze Medal, International Zhautykov Olympiad in Physics

⁷⁷4,000 teams (of size 3) from all over the world competed in a 24-hour time span against each other to solve a set of programming problems. Konstantin and Alibek scored high despite being just 2 on the team!

2014 (Malinovsky)	Participant, All-Russian Physics Olympiad
2014 (S. Hanzely)	1st Place, Slovak Mathematical Olympiad, Regional Round
2014 (Kovalev)	Honorable Mention, 15th Asian Physics Olympiad, Singapore
2014 (Kovalev)	Winner, All Russian Mathematics Olympiad (Moscow Region)
2014 (Kovalev)	Winner, All Russian Computer Science Olympiad (Moscow Region)
2014 (Kovalev)	Prizewinner, All Russian Physics Olympiad
2014 (Mishchenko)	3rd Prize, MIPT Student Mathematical Olympiad, Moscow, Russia
2014 (Horváth)	18th Place, National Mathematical Olympiad, Bratislava, Slovakia
2014 (Horváth)	1st Place, Nitra Region Mathematical Olympiad, Category A, Slovakia
2014 (Sailanbayev)	2nd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2014 (Loizou)	Top 1% in Mathematics at National and Kapodestrian University of Athens, Greece
2014 (Csiba)	Best Student Work in Applied Informatics in Czech and Slovak Republic, Annual Student Scientific Conference, Ústí nad Labem, Czech Republic
2014 (F. Hanzely)	2nd Prize (101st place), International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2014 (F. Hanzely)	9th Place, V. Jarník International Mathematical Competition, Ostrava, Czech Republic
2014 (Lukáček)	26th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic
2013 (Karagulyan)	Semifinalist, ACM-ICPC Programming Contest, NEERC region, Tbilisi, Georgia
2013 (Karagulyan)	2nd Prize, Mirror of William Lowell Putnam Mathematical Competition
2013 (Malinovsky)	Prizewinner, All-Russian Physics Olympiad
2013 (S. Hanzely)	1st Place, Slovak Mathematical Olympiad, Regional Round
2013 (Kovalev)	Winner, All Russian Physics Olympiad
2013 (Sailanbayev)	Silver Medal, International Mathematical Olympiad, Santa Marta, Colombia
2013 (F. Hanzely)	Bronze Medal, International Mathematical Olympiad, Santa Marta, Colombia
2013 (Karagulyan)	Honourable Mention, International Mathematical Olympiad, Santa Marta, Colombia
2013 (Sailanbayev)	1st Place, National Mathematical Olympiad, Kazakhstan
2013 (F. Hanzely)	1st Place, Slovak National Round of Mathematical Olympiad, Košice, Slovakia
2013 (Sailanbayev)	Gold Medal, International Zhaitykov Olympiad, Almaty, Kazakhstan
2013 (Lukáček)	20th Place, Vojtech Jarník International Mathematical Competition, Ostrava, Czech Republic
2012 (Karagulyan)	Honourable Mention, International Mathematical Olympiad, Mar del Plata, Argentina
2012 (Kovalev)	Prizewinner, All Russian Physics Olympiad
2012 (Lukáček)	3rd Prize, International Mathematical Competition for University Students, Blagoevgrad, Bulgaria
2012 (Mishchenko)	1st Prize, Moscow Mathematical Olympiad, Moscow, Russia
2012 (Mishchenko)	1st Prize, PhysTech International Olympiad in Mathematics
2012 (Basyoni)	Silver Medal ⁷⁸ , International Mathematical Olympiad, Mar del Plata, Argentina
2012 (Sailanbayev)	Bronze Medal, International Mathematical Olympiad, Mar del Plata, Argentina
2012 (Sailanbayev)	Silver Medal, Balkan Mathematical Olympiad, Antalya, Turkey
2012 (F. Hanzely)	Bronze Medal, Middle European Mathematical Olympiad, Solothurn, Switzerland
2012 (Csiba)	FIDE International Master in Chess
2012 (Csiba)	3rd Prize, International Mathematical Competition, Blagoevgrad, Bulgaria
2012 (Konečný)	2nd Prize, International ChaLearn Competition, One shot learning of gestures from Microsoft Kinect videos
2012 (Ferroq)	Gaspard Monge Prize “for best PhD thesis defended in France 2012 in mathematics or computer science, with significant contributions to Optimization and Operations Research”
2012 (Luo)	Google Anita Borg Scholarship, China
2012 (Lukáček)	2nd Place, International Correspondence Seminar in Mathematics (iKS)

⁷⁸Historically the first silver medal at IMO by Saudi Arabia.

2011 (Lukáček)	Bronze Medal (26th Place), Middle European Mathematical Olympiad, Varaždin, Croatia
2010 (Konečný)	Honourable Mention, International Mathematical Olympiad, Astana, Kazakhstan
2010 (Csiba)	Honourable Mention, Middle European Mathematical Olympiad, Žilina, Slovakia
2008 (Konečný)	Honourable Mention, Middle European Mathematical Olympiad, Olomouc, Czech Republic
2007–2009 (Takáč)	Winner, 3rd Place and Honourable Mention (twice), International Student Scientific Conference, Czech and Slovak Republic

9. TALKS

9.1 TALKS: SUMMARY

I have given **more than 270 research talks**⁷⁹ at conferences, workshops and seminars worldwide (Australia, Austria, Belgium, Brazil, Canada, Chile, China, Cuba, France, Germany, Greece, Hong Kong, Hungary, India, Japan, Mongolia, Morocco, Netherlands, Portugal, Russia, Saudi Arabia, Slovakia, Slovenia, Spain, Switzerland, UAE, United Kingdom, Uruguay, USA). Out of these, **≈80 are plenary talks** at conferences and workshops, **≈30 are invited PhD courses and tutorials**, **≈100 are seminar talks**, and the rest are invited and contributed conference talks. I regularly give talks at the premier international optimization conferences (each taking place once in 3 years): Int. Symposium on Mathematical Programming (Rio’06, Chicago’09, Berlin’12, Pittsburgh’15, Bordeaux’18), SIAM Conf. on Optimization (Darmstadt’11, San Diego’14, Vancouver’17, Hong Kong’20), Int. Conf. on Continuous Optimization (Ontario’07, Santiago’10, Lisbon’13, Tokyo’16, Berlin’19).

9.2 PLENARY TALKS⁸⁰

12/2026	CODEC-FM: Collaborative, Open, and DECentralized training of Foundation Models , NeurIPS 2026 Workshop, Sydney, Australia
11/2026	International Conference on Computational Optimization (ICOMP 2026) , Casablanca, Morocco
09/2026	1st Conference and Workshop on Mathematics and Artificial Intelligence , Nazarbayev University, Astana, Kazakhstan
06/2026	The 9th International Conference on Dynamics of Information Systems (DIS 2026) , Ústí nad Labem, Czech Republic
05/2026	ELLIIT Symposium on Optimization for Learning , Lund, Sweden
02/2026	Workshop on Learning from Heterogeneous Sources , Simons Institute, Berkeley, USA
11/2025	KAUST Workshop on Distributed Training in the Era of Large Models , KAUST, Thuwal, Saudi Arabia
10/2025	The 3rd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2025) , Dubrovnik, Croatia
09/2025	10th Workshop on High-Dimensional Approximation (HDA2025) , Bonn, Germany
08/2025	International Workshop on Secure and Efficient Federated Learning In Conjunction with ACM AsiaCCS 2025 (FL-AsiaCCS’25) , Hanoi, Vietnam
06/2025	3rd Vienna Workshop on Computational Optimization 2025 (VWCO25) , Vienna, Austria
04/2025	Rising Stars in AI Symposium , Thuwal, KAUST, Saudi Arabia
10/2024	International Conference on Computational Optimization (ICOMP-2024) , Innopolis, Russian Federation

⁷⁹All my talks are listed on https://richtarik.org/i_talks.html

⁸⁰For the purpose of this CV, a plenary talk is any talk not given to a sub-audience; or a talk explicitly labeled as a plenary/keynote talk by the organizers of the workshop/conference. I am excluding here talks at events I organized or co-organized and declined invites to deliver a plenary talk. I am including past talks, and accepted invites to give a talk.

09/2024 **2nd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2024)**, Valencia, Spain (virtual)

08/2024 **ALGOPT2024 workshop on Algorithmic Optimization: Tools for AI and Data Science**, Louvain-la-Neuve, Belgium. Celebration of Yurii Nesterov’s 50 years long research career in optimization.

06/2024 **Federated Learning for Computer Vision (FedVision) Workshop**, CVPR, Seattle, USA (virtual)

06/2024 **Applied Algorithms for Machine Learning – a Workshop on the Future of Computation**, Paris, France

06/2024 **One World Optimization Seminar in Vienna**, Erwin Schrödinger International Institute for Mathematics and Physics, Vienna, Austria

04/2024 **Workshop on Nonsmooth Optimization and Applications (NOPTA 2024)**, In Honor of the 75th Birthday of Boris Mordukhovich, University of Antwerp, Belgium

02/2024 **Apple Workshop on Privacy Preserving Machine Learning**, Cupertino, California, USA (virtual)

12/2023 **NeurIPS 2023 Workshop on Federated Learning in the Age of Foundation Models**, New Orleans, Louisiana, USA

07/2023 **ICML 2023 Workshop. Federated Learning and Analytics in Practice: Algorithms, Systems, Applications, and Opportunities**, Honolulu, Hawaii

07/2023 **Federated and Collaborative Learning Workshop**, Simons Institute, Berkeley, USA

07/2023 **Mathematics in Armenia: Advances and Perspectives** (80th anniversary of the foundation of the Armenian National Academy of Sciences), Yerevan, Armenia

12/2022 **Optimization in the Big Data Era**, Institute of Mathematical Sciences, National University of Singapore, Singapore, **Optimization in the Big Data Era**, Institute for Mathematical Sciences, National University of Singapore, Singapore

11/2022 **KAUST Workshop on Scientific Computing and Machine Learning**, KAUST

11/2022 **Google’s 2022 Workshop on Federated Learning and Analytics**, virtual

10/2022 **MBZUAI Workshop on Collaborative Learning: From Theory to Practice**, Abu Dhabi (invited by Michael I. Jordan)

09/2022 **CrossFL: Cross-Community Federated Learning: Algorithms, Systems and Co-designs**, workshop associated with the MLSys conference, Santa Clara, USA (virtual)

06/2022 **Mathematics of Complex Data**, KTH Royal Institute of Technology, Stockholm, Sweden

05/2022 **Workshop on Stochastic Numerics, Statistical Learning, Optimization, Approximations, with Applications**, KAUST, Saudi Arabia

04/2022 **Lagrange Workshop on Federated Learning**, Lagrange Mathematics and Computing Research Center, virtual

04/2022 **Apple’s Workshop on Privacy Preserving Machine Learning**, virtual

02/2022 **Dagstuhl Seminar**, Theory of Randomized Optimization Heuristics, 3 talks, Germany

12/2021 **NeurIPS 2021 Workshop. New Frontiers in Federated Learning: Privacy, Fairness, Robustness, Personalization and Data Ownership**, Virtual

11/2021 **KAUST-GSAI Joint Workshop on Advances in AI**, Virtual

11/2021 **Google Federated Learning and Analytics Workshop**, Virtual

07/2021 **Optimization Without Borders** (celebration of the 65th Birthday of Yurii Nesterov), Sirius University, Sochi, Russia (virtual)

04/2021 **KAUST Conference on Artificial Intelligence**, 2 keynote talks, Thuwal, Saudi Arabia

08/2020 **Workshop on Privacy Preserving Machine Learning**, Apple, Virtual Workshop

07/2020 **ICML 2020 Workshop: Beyond First Order Methods in ML Systems**, Virtual

06/2020 **Mathematics of Data Science**, Virtual Conference, United Kingdom

10/2019 **School-Conference “Approximation and Data Analysis”**, Nizhny Novgorod, Russia

09/2019 **Workshop on Optimization, Statistics and Numerical Methods**, Moscow Institute of Physics and Technology, Dolgoprudny, Russia (workshop organized around my visit to MIPT)

09/2019 **50 Years of Mathematics in Bielefeld - the (new) Unity of Mathematics**, Bielefeld, Germany

09/2019	DIMACS Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization , Rutgers University, USA
06/2019	Approximation, Sampling, and Compression in High Dimensional Problems , Isaac Newton Institute for Mathematical Sciences Program on “Approximation, Sampling and Compression in Data Science”, Cambridge University, UK
02/2019	Numerical Algorithms in Nonsmooth Optimization , Thematic Program: “Modern Maximal Monotone Operator Theory: From Nonsmooth Optimization to Differential Inclusions”, Erwin Schrödinger International Institute for Mathematics and Physics (ESI), Vienna, Austria
11/2018	Statistics and Data Science Workshop , KAUST, Thuwal, KSA
09/2018	Randomized Numerical Linear Algebra and Applications , Program: Data Science, Simons Institute, Berkeley, USA
08/2018	DIMACS/TRIPODS Workshop: Optimization in Machine Learning , Lehigh University, Bethlehem, USA
07/2018	XII Brazilian Workshop on Continuous Optimization , Foz do Iguaçu, Brazil
10/2017	Optimization at Work ⁸¹ , Moscow Institute of Physics and Technology, Moscow, Russia
09/2017	Workshop on Decentralized Machine Learning, Optimization and Privacy , Lille, France
07/2017	Workshop on Convex Optimization and Applications , Fields Institute, Toronto, Canada (in honour of 70th birthday of Arkadi Nemirovski)
04/2017	Visual Computing - Modeling and Reconstruction , KAUST, Thuwal, KSA
01/2017	2017 BASP Frontiers Workshop , Villars-sur-Ollon, Switzerland
11/2016	Workshop on Distributed Machine Learning , Telecom ParisTech, Paris, France
11/2016	SIAM Warwick Student Chapter Conference on Machine Learning and Statistics , Coventry, UK
10/2016	41st Woudschoten Conference , Zeist, Netherlands. Two keynote lectures in the stream “Numerical methods for big data analytics”
09/2016	Linear Algebra and Parallel Computing at the Heart of Scientific Computing , a joint event of the Royal Society of Edinburgh and the French Embassy in London, Edinburgh, UK
09/2016	“OR58”: The 58th Annual Conference of the Operational Research Society , Portsmouth, UK (closing plenary)
06/2016	2016 Int. Workshop on Modern Optimization and Applications (MOA 2016) , Beijing, China
04/2016	Einstein Center Mathematical Colloquium “Sparsity: Statistics, Optimization, and Applications” , Berlin, Germany. “The purpose of this biannual scientific colloquium is bringing together mathematicians, scientists, and engineers to enjoy a series of talks on one topical issue of current or emerging interest to several fields within mathematics.”
03/2016	Computationally and Statistically Efficient Inference for Complex Large-scale Data , Oberwolfach, Germany
09/2015	Statistical and Computational Challenges in Large-Scale Data Analysis , Alan Turing Institute Workshop, Cambridge, UK
09/2015	LMS Inverse Day: Large-Scale and Nonlinear Inverse Problems , Edinburgh, UK
04/2015	Maxwell Institute Probability Day , Edinburgh, UK
01/2015	Optimization and Statistical Learning , Les Houches, France
01/2015	Theory of Big Data Science , University College London, UK
12/2014	Optimization Workshop, Foundations of Computational Mathematics , Montevideo, Uruguay
11/2014	46th Conference of Slovak Mathematicians , Jasná, Slovakia
09/2014	Mathematical Methods in Economics and Engineering , Smolenice, Slovakia
07/2014	Understanding Complex and Large Industrial Data , Lancaster, UK
05/2014	9th Int. Conf. on Intelligent Systems: Theories and Applications , Rabat, Morocco

⁸¹This event was organized in my honour.

02/2014	Stochastic Gradient Methods , Inst. for Pure and Applied Mathematics, Los Angeles, USA
12/2013	NeurIPS Workshop on Optimization in Machine Learning , Lake Tahoe, USA. Past plenary speakers: D. Bertsekas, L. Bottou, S. Wright (2008), N. Srebro, L. Vandenberghe, A. Nemirovski (2009), M. Schmidt, Yu. Nesterov (2010), B. Recht, S. Boyd (2011), P. Parillo, F. Bach (2012)
11/2013	International Conference on Information Technologies and Society , Slovenia
10/2013	Parallel and Distributed Algorithms for Inference and Optimization , Simons Institute for the Theory of Computing, University of California, Berkeley, USA
05/2013	Big Data Mining , Imperial College London, UK
03/2013	Fête Parisienne in Computation, Inference and Optimization , IHES, Paris, France
03/2013	Edinburgh SIAM Student Chapter Conference , Edinburgh, UK
02/2013	Big Data and Social Media , Glasgow, UK
01/2013	Optimization and Statistical Learning , Les Houches, France
07/2012	Optimization in Machine Learning , ICML workshop, Edinburgh, UK
07/2011	Optimization Workshop, Foundations of Comp. Mathematics , Budapest, Hungary
05/2011	Computational Complexity Challenges in Optimization , Edinburgh, UK

9.3 INVITED LECTURE SERIES, TUTORIALS & SUMMER SCHOOL COURSES

01/2026	Federated Optimization Tutorial (3.5 hours), Bootcamp, Simons Institute Program on Federated and Collaborative Learning, Berkeley, USA
11/2025	Introduction to Machine Learning 2 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
10/2025	Introduction to Machine Learning 1 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
07/2025	The Machine Learning Summer School (MLSS 2025) , MBour, Senegal
07/2025	Introduction to Optimization 2 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
06/2025	Introduction to Optimization 1 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
12/2024	Optimization for Machine Learning (three lectures), Applied Mathematics School, KAUST, Saudi Arabia
11/2024	Optimization for Machine Learning (five 90 min lectures), Beijing Institute for Mathematical Sciences and Applications (BIMSA), Beijing, China
02/2024	Optimization for Machine Learning (three 50 min lectures), AMCS-STAT School, KAUST, Saudi Arabia
03/2024	Machine Learning Summer School, Okinawa, Japan (declined due to a clash with annual leave)
07/2023	Eastern European Machine Learning Summer School (EEML 2023), Košice, Slovakia
06/2023	Introduction to Machine Learning 2 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
06/2023	Introduction to Machine Learning 1 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
07/2023	Eastern European Machine Learning Summer School , Košice, Slovakia
11/2022	Introduction to Optimization 2 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
11/2022	Introduction to Optimization 1 (MS course, 28 hours), Dhahran, Saudi Aramco, Saudi Arabia
06/2022	Introduction to Stochastic Gradient Descent Methods (PhD course, 22.5 hours), School of Mathematics, Physics and Informatics, Bratislava, Slovakia
06/2022	Introduction to Stochastic Gradient Descent Methods (PhD course, 18 hours), Vienna Graduate School for Computational Optimization (VGSCO), Vienna, Austria

10/2019	A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods (Mini-course, 2.5 hours), School-Conference “Approximation and Data Analysis”, Nizhny Novgorod, Russia
09/2019	A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods (Mini-course, 5 hours), Moscow Institute of Physics and Technology, Dolgoprudny, Russia
08/2019	A Guided Walk Through the ZOO of Stochastic Gradient Descent Methods (Summer School Lectures, 6 hours), International Conference on Continuous Optimization (ICCOPT 2019), Berlin, Germany
02/2019	Randomized Optimization Methods (PhD Course, 4.5 hours), Erwin Schrödinger International Institute for Mathematics and Physics (ESI), Vienna, Austria
06/2018	Stochastic Reformulations in Linear Algebra and Optimization (Summer School, 2 hours), Control, Information and Optimization, Voronovo, Moscow Region, Russia
04/2018	Introduction to Optimization for Machine Learning (short outreach course for selected Saudi university students who previously participated in the Saudi National Mathematical Olympiad or IMO, 4.5 hours), KAUST, Thuwal, KSA
08/2017	Randomized Optimization Methods (Summer School, 5 hours), Data Science Summer School (DS ³), École Polytechnique, France. Other courses: Joshua Bengio (Montreal), Deep Learning; Pradeep Ravikumar (CMU), Graphical Models; Csaba Szepesvári (Alberta/Google DeepMind), Bandits
10/2015	Randomized Methods for Big Data: From Linear Systems to Optimization (Tutorial), IEEE International Conference on Data Science and Advanced Analytics, Paris, France
2015	Randomized Algorithms for Big Data Optimization (PhD Course, 18 hours), Graduate School in Systems, Optimization, Control and Networks – Université catholique de Louvain, Belgium
09/2015	Optimization in Machine Learning (PhD Course, 8 hours), Machine Learning Thematic Trimester, Toulouse, France
07/2015	Modern Convex Optimization Methods for Large-Scale Empirical Risk Minimization (Tutorial, 2 hours, joint with M. Schmidt), ICML 2015, Lille, France
06/2014	Randomized Coordinate Descent Methods (PhD Course, 6 hours), Khronos-Persyval Days “High-Dimensional Learning and Optimization”, Grenoble, France
06/2014	Coordinate Descent Methods (Lecture, 2 hours), NATCOR PhD Course on Convex Optimization, Edinburgh, UK
02/2014	Gradient Methods for Big Data (Tutorial, 3 hours), Big Data: Challenges and Applications, Imperial College London, UK

9.4 TALKS @ RESEARCH SEMINARS

2026	University of Vienna
2025	Kempelen Institute of Intelligent Technologies, The Hong Kong Polytechnic University, Federated Learning One World Seminar (virtual), Bielefeld University, KAUST (3)
2024	Tsinghua University, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), Peking University, KAUST (2)
2023	CMOR Special Lecture @ Rice University, Qualcomm AI Seminar, Apple, Kempelen Institute of Intelligent Technologies, Slovak Academy of Sciences (2)
2022	Machine Learning NeEDS Mathematical Optimization (virtual), Federated Learning One World Seminar, KAUST (3), Better AI Meetup Bratislava, Hong Kong Baptist University, One World Seminar Series on the Mathematics of Machine Learning (virtual)
2021	University of Tartu (virtual), Portsmouth (virtual), Kempelen Institute for Intelligent Technologies, Comenius University, MBZUAI (virtual), All Russian Seminar on Optimization (virtual), Federated Learning One World Seminar (virtual; 2), KAUST (3)
2020	ESET, Optimization One World Seminar, Montréal MLOpt Seminar
2019	Huawei

2018	Bratislava, KAUST (2), Warwick, Edinburgh (2)
2017	Imperial College London, KAUST, Plymouth, Cardiff
2016	Cambridge, Edinburgh (3), Stanford (2), KAUST, The Alan Turing Institute, LSE, Southampton, Skoltech, Yandex
2015	Louvain, Oxford, IST Austria, UC Davis, UC Berkeley, Edinburgh
2014	Moscow, Paris, Hong Kong, Edinburgh (3)
2013	UC Berkeley, Google, SAS Inc, Louvain, Edinburgh
2012	Wisconsin, Cambridge, Glasgow, Cardiff, Bratislava
2011	Edinburgh, Oxford, London, Heriot-Watt, Louvain
2010	Birmingham, Nottingham, Southampton
2009	ETH Zürich, Linz, Louvain, Edinburgh (2)
2008	Liège, Bratislava
2007	Cornell (2), Louvain (2)

10. TEACHING⁸²

KAUST	Fall 2026	Stochastic Gradient Descent Methods (CS 331)
	Fall 2025	Introduction to Machine Learning (DSA 008)
	Fall 2025	Federated Learning (CS 332)
	Fall 2025	Stochastic Gradient Descent Methods (CS 331)
	Fall 2025	CS Graduate Seminar (CS 398)
	Summer 2025	Introduction to Optimization (DSA 211)
	Spring 2025	Stochastic Gradient Descent Methods* (CS 331)
	Spring 2024	Stochastic Gradient Descent Methods* (CS 331)
	Spring 2023	Federated Learning* (CS 332)
	Fall 2022	Stochastic Gradient Descent Methods* (CS 331)
	Spring 2022	Federated Learning* (CS 332)
	Fall 2021	Stochastic Gradient Descent Methods* (CS 331)
	Spring 2021	Federated Learning* (CS 332)
	Fall 2020	Stochastic Gradient Descent Methods* (CS 331)
	Spring 2020	Federated Learning* (CS 390T)
	Spring 2019	Contemporary Topics in Machine Learning* (CS 394D)
	Spring 2018	Contemporary Topics in Machine Learning* (CS 394D)
	Fall 2019	Big Data Optimization* (CS 390FF)
	Fall 2018	Big Data Optimization* (CS 390FF)
	Fall 2017	Big Data Optimization* (CS 390FF)
Edinburgh	Spring 2017	Modern Optimization Methods for Big Data Problems*
	Spring 2016	Modern Optimization Methods for Big Data Problems*
	Fall 2012	Discrete Programming and Game Theory*
	Fall 2011	Discrete Programming and Game Theory*
	Fall 2011	Discrete Programming and Game Theory*
	Spring 2015	Optimization Methods in Finance*
	Spring 2014	Optimization Methods in Finance*
	Spring 2013	Optimization Methods in Finance*
	Spring 2012	Optimization Methods in Finance*
	Spring 2011	Optimization Methods in Finance*
	Fall 2012	Game Theory*
	Fall 2011	Game Theory*
	Fall 2010	Game Theory*
	Spring 2013	Computing and Numerics

⁸²I have proposed and developed from scratch courses marked with an asterisk. I was the instructor for all courses marked in bold. I was a TA (teaching assistant / tutor) for all other courses.

	Fall 2010	Dynamic & Integer Programming
	Fall 2010	Mathematics for Chemical Engineers
Louvain	Spring 2009	Nonlinear Optimization (with Yu. Nesterov)
Cornell	Spring 2006	Optimization II/Nonlinear Optimization
	Summer 2005	Engineering Probability and Statistics*
	Fall 2003	Engineering Probability and Statistics
	Summer 2003	Engineering Probability and Statistics
	Spring 2004	Optimization II
	Spring 2005	Application of Game Theory and OR to IT
	Spring 2005	Topics in Linear Optimization
	Fall 2006	Combinatorial Optimization (PhD course taught by David Williamson)
Comenius	Fall 1998	Complex Analysis

11. CONFERENCE, STREAM, WORKSHOP & SEMINAR ORGANIZATION⁸³

05/2026	Optimization for Learning Workshop, Lund, Sweden (scientific committee)
11/2025	KAUST Workshop on Distributed Training in the Age of Large Language Models, Thuwal, Saudi Arabia (organizer)
10/2025	The 3rd IEEE International Conference on Federated Learning Technologies and Applications (FLTA 2025), Dubrovnik, Croatia (general chair)
06/2025	The 4th Workshop on Federated Learning for Computer Vision, CVPR 2025 (co-organizer)
07/2024	International Symposium on Mathematical Programming (ISMP) (stream co-organizer with Lin Xiao and Simon Lacoste-Julien)
02/2023	Rising Stars in AI Symposium, KAUST, Thuwal, Saudi Arabia
12/2022	Federated Learning Workshop, NeurIPS
03/2022	Rising Stars in AI Symposium, KAUST, Thuwal, Saudi Arabia
05/2021	SIAM Conference on Optimization, Virtual (member of the organizing committee)
06/2020–now	Federated Learning One World Seminar (FLOW) ⁸⁴ (founder and chair of the organizing committee)
11/2019	KAUST-Tsinghua-Industry Workshop on Advances in Artificial Intelligence, KAUST, Thuwal, Saudi Arabia
06/2019	Sparse Approximation and Sampling, The Alan Turing Institute, London
04/2019	A Short Course on Deep Learning and the Latest AI Algorithms, KAUST, Saudi Arabia. A 2-day course delivered by Xavier Bresson, NTU, Singapore
07/2018	International Symposium on Mathematical Programming, Bordeaux, France. Scientific Committee Member for stream 4a: “Machine Learning, Big Data, Cloud Computing, and Huge-Scale Optimization” (with A. d’Aspremont, O. Beaumont and S. Sra)
02/2018	Optimization and Big Data 2018, KAUST (co-organizer with M. Canini)
2017–now	All Hands Meetings on Big Data Optimization, KAUST (a weekly group research seminar)
09/2016	IMA Numerical Linear Algebra and Optimization, Birmingham, UK (co-organizing 2 mini-symposia)
12/2015	Mathematical Perspectives on Big Data, a joint meeting of the London and Edinburgh mathematical societies, celebrating 150th anniversary of the former, Edinburgh
12/2015	Theoretical and Computational Approaches to Large-Scale Inverse Problems, Edinburgh (Alan Turing Institute Scoping Workshop)
11/2015	Distributed Machine Learning and Optimization, Edinburgh (Alan Turing Institute Scoping Workshop)
05/2015	Optimization and Big Data 2015, Edinburgh (founder and co-organizer; with Z. Qu)
01/2015	International BASP Frontiers Workshop 2015, Villars-sur-Ollon, Switzerland
12/2014	Workshop: Numerical Algorithms and Intelligent Software, Edinburgh

⁸³I am excluding organized conference sessions.

⁸⁴<https://sites.google.com/view/one-world-seminar-series-flow/home>

09/2014	2 minisymposia at 4th IMA Conf. on Numerical Lin. Alg. and Optimisation, Birmingham
05/2014	Coordinate Descent Methods Symposium at the SIAM Conference on Optimization, San Diego (24 speakers)
2014–2017	All Hands Meetings on Big Data Optimization, University of Edinburgh (a weekly interdisciplinary research seminar attended by faculty, postdocs and PhD students from the Schools of Mathematics, Engineering and Informatics and Heriot-Watt University)
07/2013	Cluster Co-Chair, “Convex and Nonsmooth Optimization” at the International Conference on Continuous Optimization (ICCOPT), Lisbon, Portugal (with F. Glineur). We organized 23 invited sessions in the cluster (=70 speakers). ICCOPT is the premiere conference in continuous optimization, taking place once in 3 years. Our cluster was twice as large as the second largest cluster.
05/2013	Optimization and Big Data 2013, Edinburgh, 64 participants (founder and organizer)
05/2012	Optimization and Big Data 2012, Edinburgh, 62 participants (founder and organizer)
07/2011	2 minisymposia at 3rd IMA Conf. on Numerical Linear Algebra and Optimisation, Birmingham
07/2011	2 minisymposia at 24th Biennial Conf. on Numerical Analysis, Glasgow

12. COMMISSIONS OF TRUST

12.1 EXTERNAL ACTIVITIES

2027	Area Chair , AAAI
2026	Program Committee Member, OPT 2026: Optimization for Machine Learning (NeurIPS 2026 Workshop)
2026	Program Committee Member, International Conference on Computational Optimization (ICOMP 2026)
2026	Reviewer, Swiss National Science Foundation (SNSF) career fellowship
2026	External PhD Examiner for Amir Ali Farzin, Australian National University (advisors: Philipp Braun and Iman Shames)
2026	Senior Area Chair , ICML
2026	Area Chair , NeurIPS
2026	Area Chair , ICLR
2025	Senior Area Chair , NeurIPS
2025	Area Chair , ICML
2025	Area Chair , ICLR
2024	Distinguished Jury Member , the AGBA GenAI Innovation Series for evaluating the innovations for 15th Edition of annual Aegis Graham Bell Awards, supported by Ministry of Electronics and Information Technology, Ministry of Education, Government of India, New Delhi and Department of Science & Technology, Government of India, Country Partner Australian Trade and Investment Commission (Austrade)
2024–now	Action Editor , Journal of Machine Learning Research (JMLR)
2024	Area Chair , NeurIPS
2024	Area Chair , ICML
2024	Area Chair , ICLR
2023–now	Associate Editor , Numerische Mathematik
2023	External PhD Examiner for Lie He, EPFL (advisor: Martin Jaggi)
2023	External PhD Examiner for Othmane Marfoq, Inria Sophia Antipolis (advisor: Giovanni Neglia)
2023	Invited to serve as Area Chair for COLT 2023 (declined)
2023	Area Chair , NeurIPS
2023	Area Chair , ICML
2023	Area Chair , ICLR
2022–2024	Action Editor , Transactions on Machine Learning Research (TMLR)

2022	Area Chair , NeurIPS
2022	Area Chair , ICML
2022	Area Chair , ICLR
2021	Habilitation ⁸⁵ Committee Member for Dr. Aurélien Bellet, Inria Lille - Nord Europe, France (other committee members: Francis Bach, Kamalika Chaudhuri and Catuscia Palamidessi)
2021	Area Chair , NeurIPS, virtual
2021	Area Chair , ICML, virtual
2021–2022	Area Editor ⁸⁶ , Journal of Optimization Theory and Applications
2021	Reviewer of Hi!Paris Fellowship applications in machine learning ⁸⁷
2021	Associate Editor (declined invite), Journal of Artificial Intelligence and Machine Learning
2021–now	Research Mentor, Kempelen Institute of Intelligent Technologies, Bratislava, Slovakia
2021	Senior Program Committee Member , IJCAI, Montréal, Canada
2021	Area Chair , ICLR, Vienna, Austria
2021	Program Committee Member, RSEML 2021: Towards Robust, Secure and Efficient Machine Learning (AAAI 2021 Workshop), virtual
2020	External PhD Examiner for Axel Böhm, University of Vienna (advisor: Radu Ioan Bot)
2020	External PhD Examiner for Dmitry Grishchenko, Université Grenoble Alpes (advisors: Franck Iutzeler, Jérôme Malick, and Massih-Reza Amini)
2020	Area Chair , NeurIPS, Vancouver, Canada
2020	Expert Reviewer , ICML, Vienna, Austria
2020	Program Committee Member, ICML International Workshop on Federated Learning for User Privacy and Data Confidentiality
2020	Evaluator & Reviewer, European Commission H2020 grants
2020	Evaluator & Reviewer, European Commission ICT grants totaling 40+ million EUR
2020	Program Committee Member, International Workshop on Federated Learning for User Privacy and Data Confidentiality (IJCAI-PRICAI ⁸⁸), Yokohama, Japan
2020	Senior Program Committee Member , IJCAI-PRICAI, Yokohama, Japan
2019	Program Committee Member, NeurIPS, Vancouver, Canada
2019	Program Committee Member, AISTATS, Naha, Okinawa, Japan
2019	External PhD Examiner for Benjamin Dubois, École des Ponts, France (advisor: G. Obozinski)
2019–now	Handling Editor , Journal of Nonsmooth Analysis and Optimization
2019	Senior Program Committee Member , IJCAI, Macao, China
2019	Area Chair , ICML, Long Beach, California
2018–now	Associate Editor , Optimization Methods and Software
2018	Reviewer, Carnegie Trust, UK
2018	Program Committee Member, NeurIPS, Montreal, Canada
2018	Program Committee Member, ICML, Stockholm, Sweden
2018	Program Committee Member, ICLR, Vancouver, Canada
2017	Program Committee Member, NeurIPS, Long Beach, USA
2017	Program Committee Member, AAAI, New Orleans, USA
2017	Reviewer, ERC (European Research Council) Consolidator Grants
2016	Habilitation Examiner for Nicolas Couellan, Institut de Mathématiques de Toulouse, Université Paul Sabatier, France (other examiners: Jean-Baptiste Hiriart-Urruty (Toulouse))
2016	External PhD Examiner for Igor Colin, Télécom ParisTech, France (other examiners: Alexandre D’Aspremont (ENS) and Mikael Johansson (KTH))
2016	Guest Editor , Journal of Computational Mathematics (co-editors: Xiaojun Chen, Yuhong Dai, and Yinyu Ye)
2016	Reviewer, EPSRC Programme Grant Scheme

⁸⁵Habilitation á diriger des recherches

⁸⁶area: Optimization for Machine Learning

⁸⁷Hi!Paris is a new interdisciplinary center for research and education on AI and Data Analytics for Science, Business and Society launched by HEC Paris and Institut polytechnique de Paris (IP Paris). See www.hi-paris.fr

⁸⁸International Joint Conference on Artificial Intelligence – Pacific Rim International Conference on Artificial Intelligence

2016	External PhD Examiner for Hamid Reza Feyzmahdavian, Automatic Control Department, KTH Royal Institute of Technology, Sweden
2016	Program Committee Member, Symposium on Distributed Information Processing, Optimization, and Resource Management over Networks, IEEE Global Conference on Signal and Information Processing, Greater Washington, D.C., USA
2016	Program Committee Member, NeurIPS, Barcelona, Spain
2016	Program Committee Member, ICML, New York, USA
2016	Program Committee Member, International Conference on Internet of Things and Big Data, Rome, Italy
2015	Program Committee Member, AISTATS, San Diego, California
2015	Program Committee Member, 13th EUROPT Workshop on Advances in Continuous Optimization, Edinburgh
2015	Program Committee Member, ICML, Lille, France
2015	External DPhil Examiner for Sheng Fang, Mathematical Institute, University of Oxford, UK (internal examiner: Jared Tanner)
2015	Lead, Alan Turing Institute PhD Programme in Data Science (responsible, on behalf of the University of Edinburgh, for the development of the PhD programme, starting in 2017)
2015	Evaluator & Reviewer, EU Horizon 2020 grants totaling 36.2 million EUR
2015	Reviewer for Leverhulme Trust (2×)
2015	Reviewer for Isaac Newton Trust
2014–2020	Associate Editor , Optimization (Frontiers in Applied Mathematics and Statistics)
2014–2017	Steering Committee (representing School of Mathematics), Centre for Doctoral Training in Data Science, University of Edinburgh (£5.03m grant from EPSRC)
2013–2017	Member, EPSRC Peer Review College
2013	Evaluator & Reviewer, EU FP7 grants totaling 42.5 million EUR.
2013	Chief Editor (declined invite), Statistics, Optimization and Computing (SOIC)
2012–2014	Steering Committee (representing University of Edinburgh), Numerical Algorithms and Intelligent Software (£5m grant from EPSRC)
2011–2017	Reviewer, EPSRC
2011–2016	Faculty Advisor, SIAM Edinburgh Student Chapter

12.2 JOURNAL REVIEWING

Mathematical Programming, SIAM Journal on Optimization, SIAM Review, Foundations of Computational Mathematics, Journal of Machine Learning Research, Machine Learning, IEEE Signal Processing, Symposium on Theory of Computing, Computational Optimization and Applications, Optimization Methods and Software, SIAM Journal on Computing, European Journal of Operational Research, Central European Journal of Operational Research, Journal of Global Optimization.

12.4 SERVICE @ KAUST

2025	PhD Proposal Examiner for Olga Mashkova, Computer Science
2024	PhD Proposal Examiner for Eslam Abdelrahman, Computer Science
2022–2025	Member, SDAIA-KAUST Center of Excellence in Data Science and AI
2022–2024	Founding Member, KAUST AI Initiative
2022–2024	Member, AI Initiative Faculty Search Committee
2022	PhD Proposal Examiner for Fatimah Zohra, Computer Science
2022	MS Thesis Examiner for Fernando Zhapa Camacho, Computer Science
2021–2022	Member, AI Initiative Advisory Board
2021	PhD Proposal Examiner for Han Shao, Computer Science
2020–2021	Chair, Machine Learning Faculty Search Committee

2020	PhD Thesis Examiner for Adel Bibi, Computer Science (other examiners: Yi Ma (Berkeley), Wolfgang Heidrich (KAUST), Bernard Ghanem (KAUST))
2019–2021	Member, AI Initiative Committee
2019–2021	Faculty Sponsor, KAUST ACM Student Chapter
2019–2020	Chair, Machine Learning Faculty Search Committee
2019	PhD Proposal Examiner for Adel Bibi, Computer Science
2019	Member, Research Strategic Plan Working Group (representing CEMSE)
2018–2019	Chair, Artificial Intelligence Committee ⁸⁹
2018–now	Co-Founder, The Machine Learning Hub (with M. Canini, B. Ghanem and P. Kalnis)
2018–2019	CS Program Curriculum Committee Member
2018	CS Faculty Search Committee Member, Machine Learning
2017	PhD Proposal Examiner for Khalil Elkhail, Electrical Engineering
2017–2019	Elected Member of the Academic Council
2017–2018	Faculty Search Committee, Statistics and Computer Science
2017	Directed Research Project Evaluation Panel

12.5 SERVICE @ EDINBURGH

2016	Recruitment Panel, Chancellor’s Fellowships in “Mathematics of Data Science” and “Industrial Mathematics”
2016	Internal PhD Examiner for Zhanxing Zhu, School of Informatics, University of Edinburgh (external examiner: Manfred Opper (TU Berlin))
2015	PhD Admissions, Data Science
2015	Recruitment Panel, Lectureship in “Mathematics of Data Science”
2014–2015	Part of a small team at Edinburgh assisting with a bid for The Alan Turing Institute (UK National Data Science and AI Institute) and subsequently with organizational planning. The bid was successful and University of Edinburgh became one of 5 founding institutions of the Alan Turing Institute (with Oxford, Cambridge, UCL and Warwick).
2013–2016	PhD Admissions, OR & Optimization
2009–2015	Director of Studies and Personal Tutor
2009–2015	MSc Projects Coordinator, OR and Optimization Programme

13. PROFESSIONAL AFFILIATIONS

Association for Computing Machinery (ACM)
 Society for Industrial and Applied Mathematics (SIAM)
 Mathematical Optimization Society (MOS)
 Edinburgh Mathematical Society (EMS)
 Isaac Newton Institute for Mathematical Sciences (INIMS)
 Institute for Operations Research and Management Science (INFORMS)
 Foundations of Computational Mathematics (FoCM)
 Slovak Mathematical Society (SMS)

14. INDUSTRY INVOLVEMENT

14.1 INDUSTRY INVOLVEMENT: SUMMARY

company	paper(s)	comment
---------	----------	---------

⁸⁹I led a university-wide committee tasked by the President of KAUST to prepare a document mapping current AI activity at KAUST and suggesting a plan for building the AI initiative at KAUST in the next 5 years; we have written a 100+ page report.

Yandex Research	[249, 266]	
Sony AI	[240]	
Shanghai AI Lab	[214]	
JD Explore Academy	[196]	
Intel	[95]	
Microsoft Research	[79, 95, 193]	
IBM Research	[22, 78, 158]	
Samsung AI		
Facebook	[83, 187]	
Amazon	[49, 151]	
Google	[51, 52, 168]	co-development of Federated Learning
Barefoot Networks	[95]	
Baidu	[29]	
Western General Hospital	[11]	

In the past I have had research discussions with SAS, Twitter, Arup, British Geological Survey, Confbuzz and Scottish Financial Risk Academy.

14.2 INDUSTRY INVOLVEMENT: FEDERATED LEARNING (with Google)

Standard machine learning approaches require centralizing the training data on one machine or in a data-center. For models trained from user interaction with mobile devices, a new approach was just released by Google, a result of collaboration between Google, Jakub Konečný and myself. The new approach is called “Federated Learning”; it is described in my papers [51, 52] and two additional papers by Google.

Federated Learning enables mobile phones to collaboratively learn a shared prediction model while keeping all the training data on device, decoupling the ability to do machine learning from the need to store the data in the cloud. This goes beyond the use of local models that make predictions on mobile devices by bringing model training to the device as well. **The technology is now in use by around 1 billion Android devices.**

The CEO of Google, Sundar Pichai, [said](#):

“... we continue to set the pace in machine learning and AI research. We introduced a new technique for training deep neural networks on mobile devices called Federated Learning. This technique enables people to run a shared machine learning model, while keeping the underlying data stored locally on mobile phones.”

The new technology is described in a Google Research Blog, dated April 2017, to a lay audience. Selected media coverage: [Forbes](#), [The Verge](#), [Quartz](#), [TechRepublic](#), [ZDNet](#), [Computer Business Review](#), [Motherboard Vice](#), [Infoworld](#), [Venturebeat](#), [Engadget](#), [Tech Narratives](#), [GadgetHacks](#), [BGR](#), [AndroidAuthority](#), [AndroidHeadlines](#), [Tom’s Guide](#), [Digital Trends](#), [The Exponential View](#), [9to5google](#).

14.3. INDUSTRY INVOLVEMENT: YOUTUBE (with Google)

An excerpt from a support letter written to me by David J Harper, the Head of EMEA University Relations, Google Switzerland, for the purpose of a (successful) grant application:

“Google recognizes the contributions of Dr Richtárik’s research to the field of big data optimization. We have invited him to deliver a talk on his research on parallel and distributed coordinate descent methods in our internal Machine Learning seminar. The talk took place in Mountain View, California, in September 2013 and was televised via our teleconference facilities to Google offices around the globe. A variant of the algorithm⁹⁰ developed by Dr. Richtárik is in operation at Google in the YouTube recommendation engine.”

⁹⁰A variant of the method developed in [18, 24].

14.4 INDUSTRY INVOLVEMENT: LLM QUANTIZATION (with Yandex Research)

HIGGS (Hadamard Incoherence with Gaussian MSE-optimal GridS), a data-free method for compressing large language models, was developed in collaboration between Yandex Research, MIT, ISTA and KAUST, and is described in our paper “Pushing the limits of large language model quantization via the linearity theorem” (NAACL 2025). HIGGS requires neither calibration data nor gradient-based optimization, and models can be quantized in minutes directly on a laptop or smartphone, with no significant loss in quality. The method has been applied to Llama 3.1 and 3.2, DeepSeek and Qwen models, and Yandex uses it to prototype and accelerate product development. Yandex announced the method in a [press release](#) on 11 April 2025.

15. PUBLICATIONS

15.1 CITATION METRICS⁹¹

According to [Google Scholar](#), my works attracted more than 30,000 citations, and my h-index is 72.

15.2 CONFERENCE/JOURNAL ABBREVIATIONS

NeurIPS	Annual Conference on Neural Information Processing Systems (a leading conference in machine learning and artificial intelligence research)
ICML	International Conference on Machine Learning (a leading conference in machine learning and artificial intelligence research)
ICLR	International Conference on Learning Representations (a leading conference in machine learning and artificial intelligence research)
AISTATS	International Conference on Artificial Intelligence and Statistics
ALT	International Conference on Algorithmic Learning Theory
AAAI	Conference on Artificial Intelligence
UAI	Uncertainty in Artificial Intelligence
ECAI	European Conference on Artificial Intelligence
NAACL	Annual Conference of the Nations of the Americas Chapter of the ACL
MSML	Mathematical and Scientific Machine Learning
JMLR	Journal of Machine Learning Research
TMLR	Transactions on Machine Learning Research
ECML PKDD	European Conf. on Machine Learning and Principles & Practice of Knowledge Discovery in Databases
DistributedML	International Workshop on Distributed Machine Learning
ICCV	IEEE International Conference on Computer Vision
VMV	Vision, Modeling and Visualization
MLSP	IEEE International Workshop on Machine Learning for Signal Processing
PROMS	Springer Proceedings in Mathematics & Statistics
ICASSP	International Conference on Acoustics, Speech, and Signal Processing (world’s largest and most comprehensive technical conference focused on signal processing and its applications)
GlobalSIP	IEEE Global Conference on Signal and Information Processing
Allerton	Annual Allerton Conference on Communication, Control, and Computing
SPARS	Proceedings of Signal Processing with Adaptive Sparse Structured Representations
WACV	IEEE Winter Conference on Applications in Computer Vision
SPIE	Proceedings of the Society of Photo-Optical Instrumentation Engineers
OR	Operations Research Proceedings
SIGCOMM	ACM’s Special Interest Group on Data Communications, specializing in the field of communication and computer networks
SOSP	Workshop on AI Systems at Symposium on Operating Systems Principles
NSDI	USENIX Symposium on Networked Systems Design and Implementation

⁹¹These citations metric were extracted via Google Scholar in September 2025.

15.3 LIST OF PUBLICATIONS, PREPRINTS & TECHNICAL REPORTS

The papers are listed in reverse chronological order in terms of their appearance online. The `arXiv` identifier is mentioned for papers which are not yet published. Coauthors marked with (r) , (p) , (d) , (m) and (i) were my (r) esearch scientists, (p) ostdocs, (d) octoral students, (m) aster students and (i) nterns at the time of writing, respectively.

- (320) E. Shulgin^(d), J. K. H. Franke, D. von Rütte, T. H. Zhang, N. Ajroldi, K. Pöppel, B. Schölkopf, A. Klein, P. Richtárik, and A. Orvieto
Deriving hyperparameter scaling laws via modern optimization theory
NeurIPS 2026
- (319) I. Sokolov^(d), A. Tyurin, and P. Richtárik
Rennala-NSGD: asynchronous stochastic optimization beyond Euclidean geometry
NeurIPS 2026
- (318) E. Shulgin^(d), T. Gadaev, S. Khirirat^(p), and P. Richtárik
Understanding MARS: when scaling momentum provably helps
ICML 2026
- (317) E. Shulgin^(d), M. Awad, P. Richtárik, and E. Gorbunov
General analysis of LMO-based optimizers: beyond bounded variance
ICML 2026
- (316) I. Ilin^(d), P. Zmushko, and P. Richtárik
Super-Tuning: from activation-aware pruning to sparse fine-tuning
`arXiv:2607.09287`
- (315) L. Yuan, Y. Yang, D. Guo, P. Richtárik, and Z. Lin
SpecGradFilter: a spectral gradient filtering framework for taming federated heterogeneity
`arXiv:2607.04189`
- (314) X. Qian^(r) and P. Richtárik
Convergence analysis of Muon-type methods with inexact LMO in the degenerate case
`arXiv:2606.21581`
- (313) I. Sokolov^(d), L. Condat^(r), and P. Richtárik
SILAGE: memory-efficient, full-gradient-free nonconvex optimization for nested finite sums
`arXiv:2606.15832`
- (312) I. Ilin^(d) and P. Richtárik
Demystifying pipeline parallelism: first theory for PipeDream
`arXiv:2606.03498`
- (311) A. Sadiev^(d), L. Condat^(r), and P. Richtárik
A unified primal-dual recipe for accelerating three-operator splitting methods
`arXiv:2605.26985`

- (310) Y. Maziane^(d), A. Mahran^(d), A. Maranjyan^(p), and P. Richtárik
LOSCAR-SGD: local SGD with communication-computation overlap and delay-corrected sparse model averaging
NeurIPS 2026
- (309) Y. Demidovich^(p), A. Chakraborty, G. Malinovsky^(d), A. Nedić, and P. Richtárik
Distance-aware Muon: adaptive step scaling for normalized optimization
arXiv:2605.18999
- (308) A. Sadiev^(d), A. Maranjyan^(p), I. Ilin^(d), and P. Richtárik
Ringmaster LMO: asynchronous linear minimization oracle momentum method
arXiv:2605.18174
- (307) A. Mahran^(d), A. Maranjyan^(p), and P. Richtárik
Rescaled asynchronous SGD: optimal distributed optimization under data and system heterogeneity
NeurIPS 2026
- (306) Z. Tovmasyan^(m), A. Maranjyan^(p), and P. Richtárik
Rennala MVR: improved time complexity for parallel stochastic optimization via momentum-based variance reduction
arXiv:2605.08871
- (305) P. Richtárik, K. Grunkowska^(d), and H. Li^(d)
Local LMO: constrained gradient optimization via a local linear minimization oracle
arXiv:2605.08850
- (304) K. Grunkowska^(d), H. Li^(d), X. Qian^(r), and P. Richtárik
Broximal alignment for global non-convex optimization
arXiv:2604.13483
- (303) X. Qian^(r), A. Gaponov^(m), G. Malinovsky^(d), and P. Richtárik
Communication-efficient Gluon in federated learning
arXiv:2604.10689
- (302) L. Condat^(r), A. Sadiev^(d), and P. Richtárik
A Nesterov-accelerated primal-dual splitting algorithm for convex nonsmooth optimization
arXiv:2604.09245
- (301) H. Li^(d), K. Grunkowska^(d), and P. Richtárik
Stabilized proximal point method via trust region control
arXiv:2604.02943
- (300) R. Islamov, G. Malinovsky^(d), A. Gaponov^(m), A. Lucchi, P. Richtárik, and E. Gorbunov
Byzantine-robust and differentially private federated optimization under weaker assumptions
UAI 2026
- (299) L. Condat^(r), A. Maranjyan^(d), and P. Richtárik
BiCoLoR: Communication-efficient optimization with bidirectional compression and local training

Prepared in 2025

- (298) E. Shulgin^(d), G. Malinovsky^(d), S. Khirirat^(p), and P. Richtárik
First provable guarantees for practical private FL: beyond restrictive assumptions
arXiv:2512.21521
- (297) A. Fradin⁽ⁱ⁾, A. Sadiev^(d), L. Condat^(r), and P. Richtárik
Tight lower bounds and optimal algorithms for stochastic nonconvex optimization with heavy-tailed noise
AISTATS 2026
- (296) X. Qian, H. Rammal⁽ⁱ⁾, D. Kovalev, and P. Richtárik
Muon is provably faster with momentum variance reduction
arXiv:2512.16598
- (295) S. Khirirat^(p), A. Sadiev^(d), Y. Demidovich^(p), and P. Richtárik
Better LMO-based momentum methods with second-order information
arXiv:2512.13227
- (294) A. Sadiev^(d), Y. Demidovich^(p), I. Sokolov^(d), G. Malinovsky^(d), S. Khirirat^(p), and P. Richtárik
Improved convergence in parameter-agnostic error feedback through momentum
arXiv:2511.14501
- (293) E. Shulgin^(d), S. AlRashed, F. Orabona, and P. Richtárik
Beyond the ideal: Analyzing the inexact Muon update
AISTATS 2026
- (292) A. Sadiev^(d), P. Richtárik, and I. Fatkhullin
Second-order optimization under heavy-tailed noise: Hessian clipping and sample complexity limits
NeurIPS 2025
- (291) K. Gruntkowska^(d), Y. Maziane^(d), Z. Qu, and P. Richtárik
Drop-Muon: Update less, converge faster
arXiv:2510.02239
- (290) K. Gruntkowska^(d) and P. Richtárik
Non-Euclidean broximal point method: a blueprint for geometry-aware optimization
arXiv:2510.00823
- (289) K. Gruntkowska^(d), A. Gaponov^(m), Z. Tovmasyan^(m), and P. Richtárik
Error feedback for Muon and friends
ICLR 2026
- (288) A. Fradin⁽ⁱ⁾, P. Richtárik and A. Tyurin
Local SGD and federated averaging through the lens of time complexity
arXiv:2509.23207
- (287) A. Maranjyan^(d) and P. Richtárik
Ringleader ASGD: The first asynchronous SGD with optimal time complexity under

data heterogeneity
ICLR 2026

- (286) L. Condat^(r) and P. Richtárik
Convergence analysis of the PAGE stochastic algorithm for convex finite-sum optimization
Journal of Optimization Theory and Applications 210:33, 2026
- (285) I. Sokolov^(d), A. Sadiev^(d), Y. Demidovich^(p), F. S. Al-Qahtani, and P. Richtárik
Bernoulli-LoRA: A theoretical framework for randomized low-rank adaptation
arXiv:2508.03820
- (284) A. Riabinin^(d), E. Shulgin^(d), K. Gruntkowska^(d), P. Richtárik
Gluon: Making Muon & Scion great again! (Bridging theory and practice of LMO-based optimizers for LLMs)
arXiv:2505.13416
- (283) L. Condat^(r), E. Gasanov^(d), P. Richtárik
The stochastic multi-proximal method for nonsmooth optimization
arXiv:2505.12409
- (282) I. Ilin^(d), P. Richtárik
Thanos: A block-wise pruning algorithm for efficient large language model compression
arXiv:2504.05346
- (281) A. Beikmohammadi, S. Khirirat^(p), P. Richtárik, and S. Magnússon
Collaborative value function estimation under model mismatch: a federated temporal difference analysis
ECML PKDD 2025
- (280) K. Burlachenko^(d), P. Richtárik
BurTorch: Revisiting training from first principles by coupling autodiff, math optimization, and systems
arXiv:2503.13795
- (279) E. Shulgin^(d), S. Khirirat^(p), P. Richtárik
Smoothed normalization for efficient distributed private optimization
arXiv:2502.13482
- (278) A. Riabinin^(d), A. Khaled, P. Richtárik
A novel unified parametric assumption for nonconvex optimization
arXiv:2502.12329
- (277) R. Islamov, S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov
Double momentum and error feedback for clipping with fast rates and differential privacy
arXiv:2502.11682
- (276) Z. Tovmasyan⁽ⁱ⁾, G. Malinovsky^(d), L. Condat^(r), P. Richtárik
Revisiting stochastic proximal point methods: generalized smoothness and similarity
Journal of Nonlinear and Variational Analysis 10(3):471–505, 2026

- (275) K. Gruntkowska^(d), H. Li^(d), A. Rane⁽ⁱ⁾, P. Richtárik
The ball-proximal (=”broximal”) point method: a new algorithm, convergence theory, and applications
arXiv:2502.02002
- (274) A. Maranjyan^(d), E.M. Saad, P. Richtárik, F. Orabona
ATA: Adaptive task allocation for efficient resource management in distributed machine learning
ICML 2025
- (273) K. Yi^(d), P. Richtárik
Symmetric pruning of large language models
arXiv:2501.18980
- (272) A. Maranjyan^(d), A. Tyurin, P. Richtárik
Ringmaster ASGD: The first asynchronous SGD with optimal time complexity
ICML 2025

Prepared in 2024

- (271) E. Shulgin^(d), P. Richtárik
On the convergence of DP-SGD with adaptive clipping
arXiv:2412.19916
- (270) I. Sokolov^(d), P. Richtárik
MARINA-P: Superior performance in non-smooth federated optimization with adaptive stepsizes
arXiv:2412.17082
- (269) A. Maranjyan^(d), A. Sadiev^(d), P. Richtárik
Differentially private random block coordinate descent
arXiv:2412.17054
- (268) E. Gasanov^(d), P. Richtárik
Speeding up stochastic proximal optimization in the high Hessian dissimilarity setting
arXiv:2412.13619
- (267) Y. Demidovich^(p), P. Ostroukhov, G. Malinovsky^(d), S. Horváth, M. Takáč, P. Richtárik, E. Gorbunov
Methods with local steps and random reshuffling for generally smooth non-convex federated optimization
ICLR 2025
- (266) V. Malinovskii, A. Panferov, I. Ilin^(m), H. Guo, P. Richtárik, D. Alistarh
Pushing the limits of large language model quantization via the linearity theorem
NAACL 2025
- (265) S. Khirirat^(p), A. Sadiev^(d), A. Riabinin^(d), E. Gorbunov, P. Richtárik
Error feedback under (L_0, L_1) -smoothness: normalization and momentum
NeurIPS 2025

- (264) W. Anyszka⁽ⁱ⁾, K. Gruntkowska^(d), A. Tyurin^(p), P. Richtárik
Tighter performance theory of FedExProx
ICLR 2026
- (263) K. Burlachenko^(d), P. Richtárik
Unlocking FedNL: Self-contained compute-optimized implementation
arXiv:2410.08760
- (262) Grigory Malinovsky^(d), Umberto Michieli, Hasan Abed Al Kader Hammoud, Taha Ceritli, Hayder Elesedy, Mete Ozay, P. Richtárik
Randomized asymmetric chain of LoRA: The first meaningful theoretical framework for low-rank adaptation
arXiv:2410.08305
- (261) A. Maranjyan^(d), O. S. Omar^(m), P. Richtárik
MindFlayer: Efficient asynchronous parallel SGD in the presence of heterogeneous and random worker compute times
UAI 2025
- (260) H. Li^(d), P. Richtárik
On the convergence of FedProx with extrapolation and inexact prox
arXiv:2410.01410
- (259) E. Gorbunov, N. Tupitsa, S. Choudhury, A. Aliev, P. Richtárik, S. Horváth, M. Takáč
Methods for convex (L_0, L_1) -smooth optimization: clipping, acceleration, and adaptivity
ICLR 2025
- (258) K. Yi^(d), T. Kharisov⁽ⁱ⁾, I. Sokolov^(d), P. Richtárik
Cohort squeeze: Beyond a single communication round per cohort in cross-device federated learning
NeurIPS 2024 FL Workshop
[Oral at the NeurIPS 2024 FL Workshop](#)
- (257) G. Meinhardt⁽ⁱ⁾, K. Yi^(d), L. Condat^(r), P. Richtárik
Prune at the clients, not the server: Accelerated sparse training in federated learning
arXiv:2405.20623
- (256) A. Karagulyan^(p), E. Shulgin^(d), A. Sadiev^(d), P. Richtárik
SPAM: Stochastic proximal point method with momentum variance reduction for non-convex cross-device federated learning
arXiv:2405.20127
- (255) L. Condat^(r), P. Richtárik
A simple linear convergence analysis of the Point-SAGA algorithm
arXiv:2405.19951
- (254) P. Richtárik, S. M. Giancola⁽ⁱ⁾, D. Lubczyk⁽ⁱ⁾, R. Yadav⁽ⁱ⁾
Local curvature descent: squeezing more curvature out of standard and Polyak gradient descent
NeurIPS 2025
- (253) A. Tyurin^(p), P. Richtárik
On the optimal time complexities in decentralized stochastic asynchronous optimization

- (252) P. Richtárik, A. Sadiev^(d), Y. Demidovich^(p)
A unified theory of stochastic proximal point methods without smoothness
arXiv:2405.15941
- (251) I.-V. Modoranu, M. Safaryan, G. Malinovsky^(d), E. Kurtic, T. Robert, P. Richtárik, D. Alistarh
MicroAdam: Accurate adaptive optimization with low space overhead and provable convergence
NeurIPS 2024
- (250) A. Tyurin^(p), K. Gruntkowska^(d), P. Richtárik
Freya PAGE: First optimal time complexity for large-scale nonconvex finite-sum optimization with heterogeneous asynchronous computations
NeurIPS 2024
- (249) V. Malinovskii, D. Mazur, I. Ilin^(d), D. Kuznedelev, K. Burlachenko^(d), K. Yi^(d), D. Alistarh, P. Richtárik
PV-Tuning: Beyond straight-through estimation for extreme LLM compression
NeurIPS 2024
Oral at NeurIPS 2024 (0.4% acceptance rate)
- (248) A. Sadiev^(d), L. Condat^(r), P. Richtárik
Stochastic proximal point methods for monotone inclusions under expected similarity
arXiv:2405.14255
- (247) H. Li^(d), K. Acharya⁽ⁱ⁾, P. Richtárik
The power of extrapolation in federated learning
NeurIPS 2024
- (246) Kai Yi^(d), G. Meinhardt⁽ⁱ⁾, P. Richtárik
FedComLoc: Communication-efficient distributed training of sparse and quantized models
arXiv:2403.09904
- (245) Y. Demidovich^(p), G. Malinovsky^(d), P. Richtárik
Streamlining in the Riemannian realm: Efficient Riemannian optimization with loopless variance reduction
arXiv:2403.06677
Best Paper Award (runner-up), International Conference on Computational Optimization (ICOMP 2025)
- (244) L. Condat^(r), A. Maranjyan^(d), P. Richtárik
LoCoDL: Communication-efficient distributed learning with local training and compression
ICLR 2025
Spotlight at ICLR 2025
- (243) K. Gruntkowska^(d), A. Tyurin^(p), P. Richtárik
Improving the worst-case bidirectional communication complexity for nonconvex distributed optimization under function similarity
NeurIPS 2024

Spotlight at NeurIPS 2024

- (242) A. Tyurin^(p), M. Pozzi⁽ⁱ⁾, I. Ilin^(d), P. Richtárik
Shadowheart SGD: distributed asynchronous SGD with optimal time complexity under arbitrary computation and communication heterogeneity
NeurIPS 2024
- (241) A. Panferov⁽ⁱ⁾, Y. Demidovich^(p), A. Rammal⁽ⁱ⁾, P. Richtárik
Correlated quantization for faster nonconvex distributed optimization
UAI 2025
Oral at UAI 2025

Prepared in 2023

- (240) K. Yi^(d), N. Gazagnadou, P. Richtárik, and L. Lyu
FedP3: Personalized and privacy-friendly federated network pruning under model heterogeneity
ICLR 2024
- (239) P. Richtárik, E. Gasanov^(d), and K. Burlachenko^(d)
Error feedback reloaded: From quadratic to arithmetic mean of smoothness constants
ICLR 2024
- (238) J. Xin, I. Ilin^(d), S. Zhang, M. Canini, and P. Richtárik
Kimad: Adaptive gradient compression with bandwidth awareness
DistributedML 2023
- (237) K. Burlachenko^(d), A. Alrowithi, F. A. Albalawi, and P. Richtárik
Federated learning is better with non-homomorphic encryption
DistributedML 2023
- (236) Y. Demidovich^(p), G. Malinovsky^(d), E. Shulgin^(d), and P. Richtárik
MAST: Model-agnostic sparsified training
ICLR 2025
- (235) G. Malinovsky^(d), P. Richtárik, S. Horváth, and E. Gorbunov
Byzantine robustness and partial participation can be achieved simultaneously: just clip gradient differences
arXiv:2311.14127
- (234) M. Fornasier, K. Riedl, P. Richtárik, and L. Sun^(d)
Consensus-based optimization with truncated noise
European Journal of Applied Mathematics, 2024
- (233) A. Rammal⁽ⁱ⁾, K. Grutkowska⁽ⁱ⁾, N. Fedin, E. Gorbunov, and P. Richtárik
Communication compression for Byzantine robust learning: New efficient algorithms and improved rates
AISTATS 2024
- (232) H. Li^(d), A. Karagulyan^(p), and P. Richtárik
MARINA meets matrix stepsizes: Variance reduced distributed non-convex optimization

arXiv:2310.04614

- (231) E. Gorbunov, A. Sadiev^(d), M. Danilova, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov, and P. Richtárik
High-probability convergence for composite and distributed stochastic minimization and variational inequalities with heavy-tailed noise
ICML 2024
Oral (144/9473 = top 1.5%)
- (230) E. Shulgin^(d) and P. Richtárik
Towards a better theoretical understanding of independent subnetwork training
ICML 2024
- (229) R. Szlendak^(m), E. Gasanov^(d), and P. Richtárik
Understanding progressive training through the framework of randomized coordinate descent
AISTATS 2024
- (228) M. Grudzień, G. Malinovsky^(d), and P. Richtárik
Improving accelerated federated learning with compression and importance sampling
arXiv:2306.03240
- (227) S. Khirirat^(p), E. Gorbunov, S. Horváth, R. Islamov, F. Karay, and P. Richtárik
Clip21: Error feedback for gradient clipping
arXiv:2305.18929
- (226) J. Xin, M. Canini, P. Richtárik, and S. Horváth
Quantize once, train fast: allreduce-compatible compression with provable guarantees
ECAI 2025
- (225) Y. Demidovich^(p), G. Malinovsky^(d), I. Sokolov^(d) and P. Richtárik
A guide through the zoo of biased SGD
NeurIPS 2023
- (224) P. Richtárik, E. Gasanov^(d) and K. Burlachenko^(d)
Error feedback shines when features are rare
arXiv:2305.15264
- (223) I. Fatkhullin, A. Tyurin^(p), and P. Richtárik
Momentum provably improves error feedback!
NeurIPS 2023
- (222) K. Yi^(d), L. Condat^(r), and P. Richtárik
Explicit personalization and local training: double communication acceleration in federated learning
TMLR 2025
- (221) A. Tyurin^(p) and P. Richtárik
Optimal time complexities of parallel stochastic optimization methods under a fixed computation model
NeurIPS 2023

- (220) A. Tyurin^(p) and P. Richtárik
2Direction: Theoretically faster distributed training with bidirectional communication compression
NeurIPS 2023
- (219) H. Li^(d), A. Karagulyan^(p) and P. Richtárik
Det-CGD: Compressed gradient descent with matrix stepsizes for non-convex optimization
ICLR 2024
- (218) A. Karagulyan^(p) and P. Richtárik
ELF: Federated Langevin algorithms with primal, dual and bidirectional compression
UAI 2025
- (217) L. Condat^(r), G. Malinovsky^(d), and P. Richtárik
TAMUNA: Accelerated federated learning with local training and partial participation
arXiv:2302.09832
- (216) G. Malinovsky^(d), S. Horváth, K. Burlachenko^(d) and P. Richtárik
Federated learning with regularized client participation
arXiv:2302.03662
- (215) A. Sadiev^(d), M. Danilova, E. Gorbunov, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov and P. Richtárik
High-probability bounds for stochastic optimization and variational inequalities: the case of unbounded variance
ICML 2023
- (214) X. Qian^(p), H. Dong, T. Zhang and P. Richtárik
Catalyst acceleration of error compensated methods leads to better communication complexity
AISTATS 2023
- (213) S. Hanzely^(d), K. Mishchenko^(d) and P. Richtárik
Convergence of first-order algorithms for meta-learning with Moreau envelopes
arXiv:2301.06806

Prepared in 2022

- (212) M. Grudzień⁽ⁱ⁾, G. Malinovsky^(d) and P. Richtárik
Can 5th generation local training methods support client sampling? Yes!
AISTATS 2023
- (211) M. Makarenko, E. Gasanov^(d), R. Islamov⁽ⁱ⁾, A. Sadiev^(d) and P. Richtárik
Adaptive compression for communication-efficient distributed training
TMLR 2023
- (210) S. Hanzely^(d), D. Kamzolov, D. Pasechnyuk, A. Gasnikov, P. Richtárik and M. Takáč
A damped Newton method achieves global $O(1/k^2)$ and local quadratic convergence rate
NeurIPS 2022

- (209) A. Maranjyan⁽ⁱ⁾, M. Safaryan^(p) and P. Richtárik
GradSkip: Communication-accelerated local gradient methods with better computational complexity
TMLR 2025
- (208) L. Condat^(r), I. Agarský^(d) and P. Richtárik
CompressedScaffnew: the first theoretical double acceleration of communication from local training and compression in distributed optimization
Optimization, 2026
- (207) L. Sun^(d) and P. Richtárik
Improved Stein variational gradient descent with importance weights
arXiv:2210.00462
- (206) K. Gruntkowska⁽ⁱ⁾, A. Tyurin^(p) and P. Richtárik
EF21-P and friends: Improved theoretical communication complexity for distributed optimization with bidirectional compression
ICML 2023
- (205) S. Bouchrouite, G. Malinovsky^(d), P. Richtárik and E. H. Bergou
Minibatch stochastic three points method for unconstrained smooth minimization
AAAI 2024
- (204) E. H. Bergou^(r), K. Burlachenko^(d), A. Dutta and P. Richtárik
Personalized federated learning with communication compression
TMLR 2023
- (203) S. Horváth^(d), K. Mishchenko^(d) and P. Richtárik
Adaptive learning rates for faster stochastic gradient methods
arXiv:2208.05287
- (202) L. Condat^(r) and P. Richtárik
RandProx: Primal-dual optimization algorithms with randomized proximal updates
ICLR 2023
OPT2022: 14th Annual Workshop on Opt. for Machine Learning (NeurIPS 2022 Workshop)
- (201) G. Malinovsky^(d), K. Yi^(d) and P. Richtárik
Variance reduced ProxSkip: Algorithm, theory and application to federated learning
NeurIPS 2022
- (200) A. Sadiev⁽ⁱ⁾, D. Kovalev^(d) and P. Richtárik
Communication acceleration of local gradient methods via an accelerated primal-dual algorithm with inexact prox
NeurIPS 2022
- (199) E. Shulgin^(d) and P. Richtárik
Shifted compression framework: generalizations and improvements
UAI 2022
- (198) L. Sun^(d) and P. Richtárik
A note on the convergence of mirrored Stein variational gradient descent under (L_0, L_1) smoothness condition

- (197) A. Sadiev⁽ⁱ⁾, G. Malinovsky^(d), E. Gorbunov, I. Sokolov^(d), A. Khaled, K. Burlachenko^(d) and P. Richtárik
Don't compress gradients in random reshuffling: compress gradient differences
NeurIPS 2024
- (196) R. Islamov⁽ⁱ⁾, X. Qian^(p), S. Hanzely^(d), M. Safaryan^(p) and P. Richtárik
Distributed Newton-type methods with communication compression and Bernoulli aggregation
TMLR 2023
NeurIPS Workshop 2022 (Order up! The Benefits of Higher-Order Optimization in Machine Learning)
- (195) M. Alfarra, J. C. Pérez, E. Shulgin^(d), P. Richtárik and B. Ghanem
Certified robustness in federated learning
NeurIPS Workshop 2022 (Federated Learning)
- (194) A. Tyurin^(p), L. Sun^(d), K. Burlachenko^(d) and P. Richtárik
Sharper rates and flexible framework for nonconvex SGD with client and data sampling
TMLR 2023
- (193) L. Sun^(d), A. Salim and P. Richtárik
Federated sampling with Langevin algorithm under isoperimetry
TMLR 2024
- (192) E. Gorbunov, S. Horváth^(d), P. Richtárik and G. Gidel
Variance reduction is an antidote to Byzantines: better rates, weaker assumptions and communication compression as a cherry on the top
ICLR 2023
- (191) L. Sun^(d), A. Karagulyan^(p) and P. Richtárik
Convergence of Stein variational gradient descent under a weaker smoothness condition
AISTATS 2023
- (190) A. Tyurin^(p) and P. Richtárik
A computation and communication efficient method for distributed nonconvex problems in the partial participation setting
NeurIPS 2023
- (189) L. Condat^(r), K. Yi^(d) and P. Richtárik
EF-BV: A unified theory of error feedback and variance reduction mechanisms for biased and unbiased compression in distributed optimization
NeurIPS 2022
- (188) G. Malinovsky^(d) and P. Richtárik
Federated random reshuffling with compression and variance reduction
arXiv:2205.03914
- (187) S. Horváth^(d), M. Sanjabi, L. Xiao, P. Richtárik and M. Rabbat
FedShuffle: Recipes for better use of local work in federated learning
TMLR 2022

- (186) K. Mishchenko^(d), G. Malinovsky^(d), S. Stich and P. Richtárik
ProxSkip: Yes! Local gradient steps provably lead to communication acceleration! Finally!
ICML 2022
- (185) D. Kovalev^(d), A. Beznosikov, A. Sadiev, M. Pershianov, P. Richtárik and A. Gasnikov
Optimal algorithms for decentralized stochastic variational inequalities
NeurIPS 2022
- (184) A. Tyurin^(p) and P. Richtárik
DASHA: Distributed nonconvex optimization with communication compression and optimal oracle complexity
ICLR 2023
- (183) P. Richtárik, I. Sokolov^(m), I. Fatkhullin⁽ⁱ⁾, E. Gasanov^(d), Z. Li^(r) and E. Gorbunov
3PC: Three point compressors for communication-efficient distributed training and a better theory for lazy aggregation
ICML 2022
- (182) H. Zhao, B. Li, Z. Li^(r), P. Richtárik and Y. Chi
BEER: Fast $O(1/T)$ rate for decentralized nonconvex optimization with communication compression
NeurIPS 2022
- (181) G. Malinovsky^(d), K. Mishchenko^(d) and P. Richtárik
Server-side stepsizes and sampling without replacement provably help in federated optimization
DistributedML 2023

Prepared in 2021

- (180) D. Kovalev^(d), A. Gasnikov and P. Richtárik
Accelerated primal-dual gradient method for smooth and convex-concave saddle-point problems with bilinear coupling
NeurIPS 2022
- (179) H. Zhao, K. Burlachenko^(d), Z. Li^(r) and Peter Richtárik
Faster rates for compressed federated learning with client-variance reduction
 To appear in: *SIAM Journal on Mathematics of Data Science, 2023*
 arXiv:2112.13097
- (178) K. Burlachenko^(d), S. Horváth^(d) and P. Richtárik
FL-PyTorch: Optimization research simulator for federated learning
DistributedML 2021
- (177) E. Gasanov^(d), A. Khaled, S. Horváth and P. Richtárik
FLIX: A simple and communication-efficient alternative to local methods in federated learning
AISTATS 2022
- (176) X. Qian^(p), R. Islamov⁽ⁱ⁾, M. Safaryan^(p) and P. Richtárik
Basis matters: better communication-efficient second order methods for federated

learning
AISTATS 2022

- (175) A. Beznosikov, P. Richtárik, M. Diskin, M. Ryabinin and A. Gasnikov
Distributed methods with compressed communication for solving variational inequalities, with theoretical guarantees
NeurIPS 2022
- (174) Rafał Szlendak^(d), A. Tyurin^(p) and P. Richtárik
Permutation compressors for provably faster distributed nonconvex optimization
ICLR 2022
- (173) I. Fatkhullin⁽ⁱ⁾, I. Sokolov^(d), E. Gorbunov^(d), Z. Li^(p) and P. Richtárik
EF21 with bells & whistles: practical algorithmic extensions of modern error feedback
JMLR 26(189):1–50, 2025
- (172) X. Qian^(p), H. Dong, P. Richtárik and T. Zhang
Error compensated loopless SVRG, Quartz, and SDCA for distributed optimization
arXiv:2109.10049
- (171) M. Jahani, S. Rusakov, Z. Shi, P. Richtárik, M. W. Mahoney and M. Takáč
Doubly adaptive scaled algorithm for machine learning using second-order information
ICLR 2022
- (170) H. Zhao, Z. Li^(r) and P. Richtárik
FedPAGE: A fast local method for federated learning
arXiv:2108.04755
- (169) Z. Li^(r) and P. Richtárik
CANITA: Faster rates for distributed convex optimization with communication compression
NeurIPS 2021
- (168) 50+ authors
A field guide to federated optimization
arXiv:2107.06917
- (167) P. Richtárik, I. Sokolov^(m), and I. Fatkhullin⁽ⁱ⁾
EF21: A new, simpler, theoretically better, and practically faster error feedback
NeurIPS 2021
NeurIPS 2021 oral paper (less than 1% acceptance rate)
- (166) D. Kovalev^(d), E. Gasanov^(d), P. Richtárik, and A. Gasnikov
Lower bounds and optimal algorithms for smooth and strongly convex decentralized optimization over time-varying networks
NeurIPS 2021
- (165) B. Wang⁽ⁱ⁾, M. Safaryan^(p), and P. Richtárik
Theoretically better and numerically faster distributed optimization with smoothness-aware quantization techniques
NeurIPS 2022

- (164) A. Salim^(p), L. Sun^(d), and P. Richtárik
A convergence theory for SVGD in the population limit under Talagrand’s inequality T1
ICML 2022
- (163) L. Condat^(r) and P. Richtárik
MURANA: A generic framework for stochastic variance-reduced optimization
MSML 2022
- (162) M. Safaryan^(p), R. Islamov⁽ⁱ⁾, X. Qian^(p), and P. Richtárik
FedNL: Making Newton-type methods applicable to federated learning
ICML 2022
- (161) G. Malinovsky^(m), A. Sailanbayev^(d), and P. Richtárik
Random reshuffling with variance reduction: new analysis and better rates
arXiv:2104.09342
- (160) Z. Li^(r) and P. Richtárik
ZeroSARAH: Efficient nonconvex finite-sum optimization with zero full gradient computations
arXiv:2103.01447
- (159) A. Salim^(p), L. Condat^(r), D. Kovalev^(d), and P. Richtárik
An optimal algorithm for strongly convex minimization under affine constraints
AISTATS 2022
- (158) Z. Shi, N. Loizou, P. Richtárik, and M. Takáč
AI-SARAH: Adaptive and implicit stochastic recursive gradient methods
TMLR 2023
- (157) D. Kovalev^(d), E. Shulgin^(m), P. Richtárik, A. Rogozin⁽ⁱ⁾, and A. Gasnikov
ADOM: Accelerated decentralized optimization method for time-varying networks
ICML 2021
- (156) K. Mishchenko^(d), B. Wang⁽ⁱ⁾, D. Kovalev^(d), and P. Richtárik
IntSGD: Floatless compression of stochastic gradients
ICLR 2022
[ICLR 2022 Spotlight paper](#)
- (155) M. Gorbunov⁽ⁱ⁾, K. Burlachenko^(d), Z. Li^(r), and P. Richtárik
MARINA: faster non-convex distributed learning with compression
ICML 2021
- (154) M. Safaryan^(p), F. Hanzely^(d), and P. Richtárik
Smoothness matrices beat smoothness constants: better communication compression techniques for distributed optimization
NeurIPS 2021
- (153) K. Islamov⁽ⁱ⁾, X. Qian^(p), and P. Richtárik
Distributed second order methods with fast rates and compressed communication
ICML 2021

- (152) K. Mishchenko^(d), A. Khaled⁽ⁱ⁾, and P. Richtárik
Proximal and federated random reshuffling
ICML 2022

Prepared in 2020

- (151) S. Horváth^(d), A. Klein, P. Richtárik, and C. Archambeau
Hyperparameter transfer learning with adaptive complexity
AISTATS 2021
- (150) X. Qian^(p), H. Dong, P. Richtárik, and T. Zhang
Error compensated loopless SVRG for distributed optimization
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (149) X. Qian^(p), H. Dong, P. Richtárik, and T. Zhang
Error compensated proximal SGD and RDA
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (148) E. Gorbunov⁽ⁱ⁾, F. Hanzely^(d), and P. Richtárik
Local SGD: unified theory and new efficient methods
AISTATS 2021
- (147) D. Kovalev^(d), A. Koloskova, M. Jaggi, P. Richtárik, and S. U. Stich
A linearly convergent algorithm for decentralized optimization: sending less bits for free!
AISTATS 2021
- (146) W. Chen⁽ⁱ⁾, S. Horváth^(d), and P. Richtárik
Optimal client sampling for federated learning
TMLR 2022
Privacy Preserving Machine Learning (NeurIPS 2020 Workshop)
- (145) E. Gorbunov⁽ⁱ⁾, D. Kovalev^(d), D. Makarenko, and P. Richtárik
Linearly converging error compensated SGD
NeurIPS 2020
- (144) A. Albasyoni^(m), M. Safaryan^(p), L. Condat^(r), and P. Richtárik
Optimal gradient compression for distributed and federated learning
SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning
- (143) F. Hanzely^(d), S. Hanzely^(m), S. Horváth^(d), and P. Richtárik
Lower bounds and optimal algorithms for personalized federated learning
NeurIPS 2020
- (142) L. Condat^(r), G. Malinovsky^(m), and P. Richtárik
Distributed proximal splitting algorithms with rates and acceleration
Frontiers in Signal Processing, section Signal Processing for Communications, 2022
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (141) R. M. Gower, M. Schmidt, F. Bach, and P. Richtárik
Variance-reduced methods for machine learning

- (140) X. Qian^(p), P. Richtárik, and T. Zhang
Error compensated distributed SGD can be accelerated
NeurIPS 2021
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (139) A. S. Berahas, M. Jahani, P. Richtárik, and M. Takáč
Quasi-Newton methods for deep learning: forget the past, just sample
Optimization Methods and Software 37(5):1668–1704, 2022
2022 Charles Broyden Prize
- (138) Z. Li^(p), H. Bao, X. Zhang, and P. Richtárik
PAGE: A simple and optimal probabilistic gradient estimator for nonconvex optimization
ICML 2021
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
Spotlight talk
- (137) D. Kovalev^(d), A. Salim^(p), and P. Richtárik
Optimal and practical algorithms for smooth and strongly convex decentralized optimization
NeurIPS 2020
- (136) Ahmed Khaled⁽ⁱ⁾, Othmane Sebbouh⁽ⁱ⁾, Nicolas Loizou, Robert M. Gower, and P. Richtárik
Unified analysis of stochastic gradient methods for composite convex and smooth optimization
arXiv:2006.11573
- (135) S. Horváth^(d) and P. Richtárik
A better alternative to error feedback for communication-efficient distributed learning
ICLR 2021
SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning
The Best Paper Award at the NeurIPS 2020 Workshop on Scalability, Privacy, and Security in Federated Learning
- (134) A. Salim^(p) and P. Richtárik
Primal dual interpretation of the proximal stochastic gradient Langevin algorithm
NeurIPS 2020
- (133) Z. Li^(p) and P. Richtárik
A unified analysis of stochastic gradient methods for nonconvex federated optimization
SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning
- (132) K. Mishchenko^(d), A. Khaled⁽ⁱ⁾, and P. Richtárik
Random reshuffling: simple analysis with vast improvements
NeurIPS 2020
- (131) M. Alfarra^(m), S. Hanzely^(m), A. Albasyoni^(m), B. Ghanem, and P. Richtárik
Adaptive learning of the optimal mini-batch size of SGD
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)

- (130) A. Salim^(p), L. Condat^(r), K. Mishchenko^(d), and P. Richtárik
Dualize, split, randomize: fast nonsmooth optimization algorithms
Journal of Optimization Theory and Applications 195:102–130, 2022
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (129) A. N. Sahu⁽ⁱ⁾, A. Dutta^(p), A. Tiwari⁽ⁱ⁾, and P. Richtárik
On the convergence analysis of asynchronous SGD for solving consistent linear systems
Linear Algebra and its Applications 663:1–31, 2023
- (128) G. Malinovsky⁽ⁱ⁾, D. Kovalev^(d), E. Gasanov^(d), L. Condat^(r), and P. Richtárik
From local SGD to local fixed point methods for federated learning
ICML 2020
- (127) A. Beznosikov⁽ⁱ⁾, S. Horváth^(d), P. Richtárik and M. Safaryan^(p)
On biased compression for distributed learning
JMLR 24(276):1–50, 2023
SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning
- (126) Z. Li^(p), D. Kovalev^(d), X. Qian^(p) and P. Richtárik
Acceleration for compressed gradient descent in distributed and federated optimization
ICML 2020
- (125) D. Kovalev^(d), R. M. Gower, P. Richtárik and A. Rogozin⁽ⁱ⁾
Fast linear convergence of randomized BFGS
arXiv:2002.11337
- (124) F. Hanzely^(d), N. Doikov, P. Richtárik and Yu. Nesterov
Stochastic subspace cubic Newton method
ICML 2020
- (123) Mher Safaryan^(p), Egor Shulgin^(m) and P. Richtárik
Uncertainty principle for communication compression in distributed and federated learning and the search for an optimal compressor
Information and Inference: A Journal of the IMA, 2021
- (122) F. Hanzely^(d) and P. Richtárik
Federated learning of a mixture of global and local models
SpicyFL 2020: NeurIPS Workshop on Scalability, Privacy, and Security in Federated Learning
- (121) S. Horváth^(d), L. Lei, P. Richtárik and M. I. Jordan
Adaptivity of stochastic gradient methods for nonconvex optimization
SIAM Journal on Mathematics of Data Science 4(2):634–648, 2022
OPT2020: 12th Annual Workshop on Optimization for Machine Learning (NeurIPS 2020 Workshop)
- (120) F. Hanzely^(d), D. Kovalev^(d) and P. Richtárik
Variance reduced coordinate descent with acceleration: new method with a surprising application to finite-sum problems
ICML 2020
- (119) A. Khaled⁽ⁱ⁾ and P. Richtárik
Better theory for SGD in the nonconvex world
TMLR 2022

Prepared in 2019

- (118) A. Khaled⁽ⁱ⁾, K. Mishchenko^(d) and P. Richtárik
Tighter theory for local SGD on identical and heterogeneous data
AISTATS 2020
- (117) S. Chraïbi⁽ⁱ⁾, A. Khaled⁽ⁱ⁾, D. Kovalev⁽ⁱ⁾, A. Salim^(p), P. Richtárik and M. Takáč
Distributed fixed point methods with compressed iterates
arXiv:1912.09925
- (116) S. Horváth^(d), C.-Y. Ho, Ľ. Horváth⁽ⁱ⁾, A. Narayan Sahu, M. Canini and P. Richtárik
IntML: Natural compression for distributed deep learning
Workshop on AI Systems at Symposium on Operating Systems Principles 2019 (SOSP'19)
- (115) D. Kovalev^(m), K. Mishchenko^(d) and P. Richtárik
Stochastic Newton and cubic Newton methods with simple local linear-quadratic rates
NeurIPS 2019 Workshop: Beyond First Order Methods in ML
- (114) A. Khaled⁽ⁱ⁾, K. Mishchenko^(d) and P. Richtárik
Better communication complexity for local SGD
NeurIPS 2019 Workshop: Federated Learning for Data Privacy and Confidentiality
- (113) A. Khaled⁽ⁱ⁾ and P. Richtárik
Gradient descent with compressed iterates
NeurIPS 2019 Workshop: Federated Learning for Data Privacy and Confidentiality
- (112) A. Khaled⁽ⁱ⁾, K. Mishchenko^(d) and P. Richtárik
First analysis of local GD on heterogeneous data
NeurIPS 2019 Workshop: Federated Learning for Data Privacy and Confidentiality
- (111) J. Xiong, P. Richtárik and W. Heidrich
Stochastic convolutional sparse coding
International Symposium on Vision, Modeling and Visualization, 2019
VMV 2019 Best Paper Award
- (110) X. Qian^(p), Z. Qu and P. Richtárik
L-SVRG and L-Katyusha with arbitrary sampling
JMLR 22(112):1–47, 2021
- (109) X. Qian^(p), A. Sailanbayev^(d), K. Mishchenko^(d) and P. Richtárik
MISO is making a comeback with better proofs and rates
arXiv:1906.01474
- (108) E. Gorbunov⁽ⁱ⁾, A. Bibi, O. Sezer, E. H. Bergou^(p) and P. Richtárik
A stochastic derivative free optimization method with momentum
NeurIPS 2019 Workshop: Optimization Foundations for Reinforcement Learning
ICLR 2020
- (107) M. Safaryan and P. Richtárik
Stochastic sign descent methods: New algorithms and better theory
ICML 2021

- (106) A. Salim^(p), D. Kovalev^(m) and P. Richtárik
Stochastic proximal Langevin algorithm: potential splitting and nonasymptotic rates
NeurIPS 2019
- (105) E. H. Bergou^(p), M. Canini, A. Dutta^(p), P. Richtárik and Y. Xiao⁽ⁱ⁾
Direct nonlinear acceleration
EURO Journal on Computational Optimization, 2022
- (104) K. Mishchenko^(d) and P. Richtárik
A stochastic decoupling method for minimizing the sum of smooth and non-smooth functions
arXiv:1905.11535
- (103) K. Mishchenko^(d), D. Kovalev^(m), E. Shulgin⁽ⁱ⁾, P. Richtárik and Y. Malitsky
Revisiting stochastic extragradient
AISTATS 2020
NeurIPS 2019 Workshop: Optimization Foundations for Reinforcement Learning
- (102) F. Hanzely^(d) and P. Richtárik
One method to rule them all: variance reduction for data, parameters and many new methods
arXiv:1905.11266
- (101) E. Gorbunov⁽ⁱ⁾, F. Hanzely^(d) and P. Richtárik
A unified theory of SGD: variance reduction, sampling, quantization and coordinate descent
AISTATS 2020
- (100) S. Horváth^(d), C.Y. Ho, L. Horváth⁽ⁱ⁾, A. N. Sahu, M. Canini and P. Richtárik
Natural compression for distributed deep learning
MSML 2022
- (99) R. M. Gower, D. Kovalev^(m), F. Lieder and P. Richtárik
RSN: Randomized Subspace Newton
NeurIPS 2019
- (98) A. Dutta^(p), F. Hanzely^(d), J. Liang and P. Richtárik
Best pair formulation & accelerated scheme for non-convex principal component pursuit
IEEE Transactions on Signal Processing 26:6128–6141, 2020
- (97) N. Loizou^(d) and P. Richtárik
Revisiting randomized gossip algorithms: general framework, convergence rates and novel block and accelerated protocols
IEEE Transactions on Information Theory 67(12):8300–8324, 2021
- (96) N. Loizou^(d) and P. Richtárik
Convergence analysis of inexact randomized iterative methods
SIAM Journal on Scientific Computing 42(6), A3979–A4016, 2020
- (95) A. Sapio, M. Canini, C.-Y. Ho, J. Nelson, P. Kalnis, C. Kim, A. Krishnamurthy, M. Moshref, D. Ports and P. Richtárik
Scaling distributed machine learning with in-network aggregation

- (94) S. Horváth^(d), D. Kovalev^(d), K. Mishchenko^(d), P. Richtárik and S. Stich
Stochastic distributed learning with gradient quantization and double variance reduction
Optimization Methods and Software 38(1):91–106, 2023
2023 Charles Broyden Prize
- (93) E. H. Bergou^(p), E. Gorbunov⁽ⁱ⁾ and P. Richtárik
Stochastic three points method for unconstrained smooth minimization
SIAM Journal on Optimization 30(4):2726–2749, 2020
- (92) E. H. Bergou^(p), A. Bibi, B. Ghanem, O. Sener and P. Richtárik
A stochastic derivative-free optimization method with importance sampling
AAAI 2020
- (91) K. Mishchenko^(d), F. Hanzely^(d) and P. Richtárik
99% of distributed optimization is a waste of time: the issue and how to fix it
UAI 2020
- (90) K. Mishchenko^(d), E. Gorbunov⁽ⁱ⁾, M. Takáč and P. Richtárik
Distributed learning with compressed gradient differences
Optimization Methods and Software 40(5):1181–1196, 2025
arXiv:1901.09269
- (89) R. M. Gower, N. Loizou^(d), X. Qian^(p), A. Sailanbayev^(d), E. Shulgin⁽ⁱ⁾ and P. Richtárik
SGD: general analysis and improved rates
ICML 2019
- (88) D. Kovalev^(d), S. Horváth^(d) and P. Richtárik
Don't jump through hoops and remove those loops: SVRG and Katyusha are better without the outer loop
ALT 2020
- (87) X. Qian^(p), Z. Qu and P. Richtárik
SAGA with arbitrary sampling
ICML 2019

Prepared in 2018

- (86) L. M. Nguyen, P. H. Nguyen, P. Richtárik, K. Scheinberg and M. Takáč and M. van Dijk
New convergence aspects of stochastic gradient algorithms
JMLR 20(176):1–49, 2019
- (85) F. Hanzely^(d), J. Konečný^(d), N. Loizou^(d), P. Richtárik and D. Grishchenko⁽ⁱ⁾
A privacy preserving randomized gossip algorithm via controlled noise insertion⁹²
NeurIPS 2018 Workshop: Privacy Preserving Machine Learning
- (84) K. Mishchenko^(d) and P. Richtárik
A stochastic penalty model for convex and nonconvex optimization with big constraints

⁹²Short version of [58]

arXiv:1810.13387

- (83) N. Loizou^(d), M. Rabbat and P. Richtárik
Provably accelerated randomized gossip algorithms
ICASSP 2019
- (82) F. Hanzely^(d) and P. Richtárik
Accelerated coordinate descent with arbitrary sampling and best rates for minibatches
AISTATS 2019
- (81) S. Horváth^(d) and P. Richtárik
Nonconvex variance reduced optimization with arbitrary sampling
ICML 2019
Horváth: Best DS³ (Data Science Summer School) Poster Award⁹³, 2018
- (80) F. Hanzely^(d), K. Mishchenko^(d) and P. Richtárik
SEGA: Variance reduction via gradient sketching
NeurIPS 2018
- (79) F. Hanzely^(d), P. Richtárik and L. Xiao
Accelerated Bregman proximal gradient methods for relatively smooth convex optimization
Computational Optimization and Applications 79:405–440, 2021
- (78) J. Mareček, P. Richtárik and M. Takáč
Matrix completion under interval uncertainty: highlights
ECML-PKDD 2018
- (77) N. Loizou^(d) and P. Richtárik
Accelerated gossip via stochastic heavy ball method
Allerton 2018 (The 56th Annual Allerton Conf. on Communication, Control, and Computing, 2018)
- (76) A. Bibi, A. Sailanbayev^(d), B. Ghanem, R. M. Gower and P. Richtárik
Improving SAGA via a probabilistic interpolation with gradient descent
arXiv:1806.05633
- (75) A. Dutta^(p), F. Hanzely^(d) and P. Richtárik
A nonconvex projection method for robust PCA
AAAI 2019
- (74) R. M. Gower, P. Richtárik and F. Bach
Stochastic quasi-gradient methods: variance reduction via Jacobian sketching
Mathematical Programming 188:135–192, 2021
- (73) A. Dutta^(p), X. Li and P. Richtárik
Weighted low-rank approximation of matrices and background modeling
arXiv:1804.06252
- (72) F. Hanzely^(d) and P. Richtárik
Fastest rates for stochastic mirror descent methods
Computational Optimization and Applications 79:717–766, 2021

⁹³The first prize out of 170 competing posters presented by MS students, PhD students and postdocs. SH is an MS student. Cash award 500 EUR.

- (71) L. M. Nguyen, P. H. Nguyen, M. van Dijk, P. Richtárik, K. Scheinberg and M. Takáč
SGD and Hogwild! convergence without the bounded gradients assumption
ICML 2018 (Proceedings of the 35th Int. Conf. on Machine Learning, PMLR 80:3750-3758, 2018)
- (70) R. M. Gower, F. Hanzely^(d), P. Richtárik and S. Stich
Accelerated stochastic matrix inversion: general theory and speeding up BFGS rules for faster second-order optimization
NeurIPS 2018
- (69) N. Doikov⁽ⁱ⁾ and P. Richtárik
Randomized block cubic Newton method
ICML 2018 (Proceedings of the 35th Int. Conf. on Machine Learning, PMLR 80:1290-1298, 2018)
Doikov: Best Talk Award at “Traditional Youth School in Control, Information and Optimization”, Voronovo, Russia, 2018
- (68) D. Kovalev⁽ⁱ⁾, E. Gorbunov⁽ⁱ⁾, E. Gasanov⁽ⁱ⁾ and P. Richtárik
Stochastic spectral and conjugate descent methods
NeurIPS 2018

- (67) R. Harman, L. Filová and P. Richtárik
A randomized exchange algorithm for computing optimal approximate designs of experiments
Journal of the American Statistical Association, 1–30, 2019
- (66) I. Necoara, A. Patrascu and P. Richtárik
Randomized projection methods for convex feasibility problems: conditioning and convergence rates
SIAM Journal on Optimization 29(4):2814–2852, 2019

Prepared in 2017

- (65) N. Loizou^(d) and P. Richtárik
Momentum and stochastic momentum for stochastic gradient, Newton, proximal point and subspace descent methods
Computational Optimization and Applications 77:653–710, 2020
- (64) A. Dutta^(p) and P. Richtárik
Online and batch supervised background estimation via L1 regression
WACV 2019
- (63) N. Loizou^(d) and P. Richtárik
Linearly convergent stochastic heavy ball method for minimizing generalization error
NeurIPS 2017 Workshop: Optimization for Machine Learning
- (62) D. Csiba^(d) and P. Richtárik
Global convergence of arbitrary-block gradient methods for generalized Polyak-Łojasiewicz functions
Submitted to: *Mathematical Programming (under 1st minor revision)*
arXiv:1709.03014

- (61) A. A. Ribeiro^(p) and P. Richtárik
The complexity of primal-dual fixed point methods for ridge regression
Linear Algebra and its Applications 556, 342-372, 2018
- (60) M. J. Ehrhardt, P. Markiewicz, A. Chambolle, P. Richtárik, J. Schott and C. B. Schönlieb
Faster PET reconstruction with a stochastic primal-dual hybrid gradient method
Proceedings of SPIE, Wavelets and Sparsity XVII, Volume 10394, pages 103941O-1 to 103941O-11, 2017
- (59) A. Dutta^(p), X. Li and P. Richtárik
A batch-incremental video background estimation model using weighted low-rank approximation of matrices
IEEE International Conference on Computer Vision (ICCV) Workshops, 2017
- (58) F. Hanzely^(d), J. Konečný^(d), N. Loizou^(d), P. Richtárik and D. Grishchenko⁽ⁱ⁾
Privacy preserving randomized gossip algorithms
arXiv:1706.07636
- (57) A. Chambolle, M.J. Ehrhardt, P. Richtárik and C.B. Schönlieb
Stochastic primal-dual hybrid gradient algorithm with arbitrary sampling and imaging applications
SIAM Journal on Optimization 28(4):2783-2808, 2018
- (56) P. Richtárik and M. Takáč
Stochastic reformulations of linear systems: algorithms and convergence theory
SIAM Journal on Matrix Analysis and Applications 41(2):487-524, 2020
- (55) M. Mutný⁽ⁱ⁾ and P. Richtárik
Parallel stochastic Newton method
Journal of Computational Mathematics 36(3):405-427, 2018

Prepared in 2016

- (54) R. M. Gower^(d) and P. Richtárik
Linearly convergent randomized iterative methods for computing the pseudoinverse
arXiv:1612.06255
- (53) J. Konečný^(d) and P. Richtárik
Randomized distributed mean estimation: accuracy vs communication
Frontiers in Applied Mathematics and Statistics 4:62, 2018
- (52) J. Konečný^(d), H. B. McMahan, F. Yu, P. Richtárik, A.T. Suresh and D. Bacon
Federated learning: strategies for improving communication efficiency
NeurIPS 2016 Worskhop: Private Multi-Party Machine Learning
- (51) J. Konečný^(d), H. B. McMahan, D. Ramage and P. Richtárik
Federated optimization: distributed machine learning for on-device intelligence
arXiv:1610.02527
- (50) N. Loizou^(d) and P. Richtárik
A new perspective on randomized gossip algorithms
GlobalSIP 2016 (The 4th IEEE Global Conference on Signal and Information Processing, 440-444,

2016)

- (49) S. J. Reddi, J. Konečný^(d), P. Richtárik, B. Póczos and A. Smola
AIDE: Fast and communication efficient distributed optimization
arXiv:1608.06879
- (48) D. Csiba^(d) and P. Richtárik
Coordinate descent face-off: primal or dual?
ALT 2018 (Proceedings of Algorithmic Learning Theory, PMLR 83:246–267, 2018)
- (47) O. Fercoq^(p) and P. Richtárik
Optimization in high dimensions via accelerated, parallel and proximal coordinate descent⁹⁴
SIAM Review 58(4), 2016
SIAM SIGEST Outstanding Paper Award, 2017
- (46) R. M. Gower^(d), D. Goldfarb and P. Richtárik
Stochastic block BFGS: squeezing more curvature out of data
ICML 2016
- (45) D. Csiba^(d) and P. Richtárik
Importance sampling for minibatches
JMLR 19(27):1–21, 2018
- (44) R. M. Gower^(d) and P. Richtárik
Randomized quasi-Newton updates are linearly convergent matrix inversion algorithms
SIAM Journal on Matrix Analysis and Applications 38(4): 1380–1409, 2017
6th Most Downloaded Paper from the SIMAX Website, 2018

Prepared in 2015

- (43) Z. Allen-Zhu, Z. Qu^(p), P. Richtárik and Y. Yuan
Even faster accelerated coordinate descent using non-uniform sampling
ICML 2016
- (42) R. M. Gower^(d) and P. Richtárik
Stochastic dual ascent for solving linear systems
arXiv:1512.06890
- (41) C. Ma, J. Konečný^(d), M. Jaggi, V. Smith, M. I. Jordan, P. Richtárik and M. Takáč
Distributed optimization with arbitrary local solvers
Optimization Methods and Software 32(4):813–848, 2017
1st Most-Read Paper in Optimization Methods and Software, 2017
- (40) M. Takáč, P. Richtárik and N. Srebro
Distributed minibatch SDCA
To appear in: *JMLR*⁹⁵

⁹⁴A (refreshed) reprint of [21] originally published in SIAM Journal on Optimization

⁹⁵We did not receive any reviews after 2.5 years since submission. The paper was recently accepted after a change in JMLR leadership.

- (39) R. M. Gower^(d) and P. Richtárik
Randomized iterative methods for linear systems
SIAM Journal on Matrix Analysis and Applications 36(4):1660-1690, 2015
 Gower: 18th Leslie Fox Prize (2nd Prize), Institute for Mathematics and its Applications, 2017
 1st Most Downloaded Paper from the SIMAX Website, 2017
 2nd Most Downloaded Paper from the SIMAX Website, 2018
 2nd Most Downloaded Paper from the SIMAX Website, 2019
 1st Most Downloaded Paper from the SIMAX Website, 2020
- (38) D. Csiba^(d) and P. Richtárik
Primal method for ERM with flexible mini-batching schemes and non-convex losses
 arXiv:1506:02227
- (37) J. Konečný^(d), J. Liu, P. Richtárik and M. Takáč
Mini-batch semi-stochastic gradient descent in the proximal setting
IEEE Journal of Selected Topics in Signal Processing 10(2):242-255, 2016
 Konečný: BASP Frontiers Best Contribution Award, 2015
- (36) R. Tappenden^(p), M. Takáč^(d) and P. Richtárik
On the complexity of parallel coordinate descent
Optimization Methods and Software 33(2), 372-395, 2018
- (35) D. Csiba^(d), Z. Qu^(p) and P. Richtárik
Stochastic dual coordinate ascent with adaptive probabilities
ICML 2015
 Csiba: Best Contribution Award (2nd Prize), Optimization and Big Data 2015
 Implemented in Tensor Flow
- (34) C. Ma, V. Smith, M. Jaggi, M. I. Jordan, P. Richtárik and M. Takáč
Adding vs. averaging in distributed primal-dual optimization
ICML 2015
 Smith: MLconf Industry Impact Student Research Award, 2015
 CoCoA+ is now the default linear optimizer in Tensor Flow
- (33) Z. Qu^(p), P. Richtárik, M. Takáč^(d) and O. Fercoq^(p)
SDNA: Stochastic dual Newton ascent for empirical risk minimization
ICML 2016

Prepared in 2014

- (32) Z. Qu^(p) and P. Richtárik
Coordinate descent with arbitrary sampling II: expected separable overapproximation
Optimization Methods and Software 31(5):858-884, 2016
 7th Most-Read Paper in Optimization Methods and Software, 2017
- (31) Z. Qu^(p) and P. Richtárik
Coordinate descent with arbitrary sampling I: algorithms and complexity
Optimization Methods and Software 31(5):829-857, 2016
 4th Most-Read Paper in Optimization Methods and Software, 2017
- (30) J. Konečný^(d), Z. Qu^(p) and P. Richtárik
Semi-stochastic coordinate descent

- (29) Z. Qu^(p), P. Richtárik and T. Zhang
Quartz: Randomized dual coordinate ascent with arbitrary sampling
NeurIPS 2015
- (28) J. Konečný^(d), J. Liu, P. Richtárik and M. Takáč^(d)
mS2GD: Mini-batch semi-stochastic gradient descent in the proximal setting⁹⁶
NeurIPS 2014 Workshop: Optimization for Machine Learning
- (27) J. Konečný^(d), Z. Qu^(p) and P. Richtárik
S2CD: Semi-stochastic coordinate descent⁹⁷
NeurIPS 2014 Workshop: Optimization for Machine Learning
- (26) J. Konečný^(d) and P. Richtárik
Simple complexity analysis of simplified direct search
 arXiv:1410.0390
- (25) J. Mareček^(p), P. Richtárik and M. Takáč^(d)
Distributed block coordinate descent for minimizing partially separable functions
PROMS 2015 (In: Al-Baali M., Grandinetti L., Purnama A. (eds) Numerical Analysis and Optimization. Springer Proceedings in Math. & Statistics, vol 134. Springer, Cham, 261–288, 2015)
- (24) O. Fercoq^(p), Z. Qu^(p), P. Richtárik and M. Takáč^(d)
Fast distributed coordinate descent for minimizing non-strongly convex losses
MLSP 2014 (2014 IEEE Int. Workshop on Machine Learning for Signal Processing, 1–6, 2014)
- (23) D. Forgan and P. Richtárik
On optimal solutions to planetesimal growth models
Technical Report ERGO 14-002, 2014
- (22) J. Mareček^(p), P. Richtárik and M. Takáč^(d)
Matrix completion under interval uncertainty
European Journal of Operational Research 256(1):35–43, 2017

Prepared in 2013

- (21) O. Fercoq^(p) and P. Richtárik
Accelerated, parallel and proximal coordinate descent
SIAM Journal on Optimization 25(4):1997–2023, 2015
 Fercoq: 17th Leslie Fox Prize (2nd Prize), Institute for Mathematics and its Applications, 2015
 2nd Most Downloaded Paper from the SIOPT Website, 2016 & 2017
- (20) J. Konečný^(d) and P. Richtárik
Semi-stochastic gradient descent
Frontiers in Applied Mathematics and Statistics 3:9, 2017
- (19) P. Richtárik and M. Takáč^(d)
On optimal probabilities in stochastic coordinate descent methods

⁹⁶A short version of the journal paper [37]

⁹⁷A short version of the journal paper [30]

Optimization Letters 10(6):1233–1243, 2016

- (18) P. Richtárik and M. Takáč^(d)
Distributed coordinate descent method for learning with big data
JMLR 17 (75):1–25, 2016
- (17) O. Fercoq^(p) and P. Richtárik
Smooth minimization of nonsmooth functions with parallel coordinate descent methods
PROMS 2017 (Modelling and Optimization: Theory and Applications, Springer Proceedings in Math. and Statistics)
- (16) R. Tappenden^(p), P. Richtárik and B. Büke
Separable approximations and decomposition methods for the augmented Lagrangian
Optimization Methods and Software 30(3):643–668, 2015
- (15) R. Tappenden^(p), P. Richtárik and J. Gondzio
Inexact coordinate descent: complexity and preconditioning
Journal of Optimization Theory and Applications 171 (1):144–176, 2016
- (14) M. Takáč^(d), S. D. Ahipasaoglu, N. M. Cheung and P. Richtárik
TOP-SPIN: TOPic discovery via Sparse Principal component INterference
PROMS 2017 (Modelling and Optimization: Theory and Applications, Springer Proceedings in Math. and Statistics)
- (13) M. Takáč^(d), A. Bijral, P. Richtárik and N. Srebro
Mini-batch primal and dual methods for SVMs
ICML 2013

Prepared in 2012 or Before

- (12) P. Richtárik, M. Takáč^(d), S. D. Ahipasaoglu and M. Jahani
Alternating maximization: unifying framework for 8 sparse PCA formulations and efficient parallel codes
Optimization and Engineering, 2020
- (11) W. Hulme^(m), P. Richtárik, L. McGuire and A. Green
Optimal diagnostic tests for sporadic Creutzfeldt-Jakob disease based on SVM classification of RT-QuIC data
Technical Report ERGO 12-014, 2012
[arXiv:1212.2617](#)
- (10) P. Richtárik and M. Takáč^(d)
Parallel coordinate descent methods for big data optimization
Mathematical Programming 156(1):433–484, 2016
[Takáč: 16th Leslie Fox Prize \(2nd Prize\)](#), Institute for Mathematics and its Applications, 2013
- (9) P. Richtárik and M. Takáč^(d)
Efficient serial and parallel coordinate descent methods for huge-scale truss topology design
In: Klatte D., Lüthi HJ., Schmedders K. (eds) Operations Research Proceedings 2011 (Gesellschaft für Operations Research e.V.). Springer, Berlin, Heidelberg, 2012

- (8) P. Richtárik and M. Takáč^(d)
Iteration complexity of randomized block-coordinate descent methods for minimizing a composite function
Mathematical Programming 144(2):1–38, 2014
 Takáč: Best Student Paper Award (sole runner-up), INFORMS Computing Society, 2012
- (7) P. Richtárik and M. Takáč^(d)
Efficiency of randomized coordinate descent methods on minimization problems with a composite objective function
SPARS 2011 (Signal Processing with Adaptive Sparse Structured Representations)
- (6) P. Richtárik
Finding sparse approximations to extreme eigenvectors: generalized power method for sparse PCA and extensions
SPARS 2011 (Signal Processing with Adaptive Sparse Structured Representations)
- (5) P. Richtárik
Approximate level method for nonsmooth convex optimization
Journal of Optimization Theory and Applications 152(2):334–350, 2012
- (4) M. Journée, Yu. Nesterov, P. Richtárik and R. Sepulchre
Generalized power method for sparse principal component analysis
JMLR 11:517–553, 2010
- (3) P. Richtárik
Improved algorithms for convex minimization in relative scale
SIAM Journal on Optimization 21(3):1141–1167, 2011
- (2) P. Richtárik
Simultaneously solving seven optimization problems in relative scale
Technical Report, Optimization Online, 2008
- (1) P. Richtárik
Some algorithms for large-scale linear and convex minimization in relative scale
PhD thesis, School of ORIE, Cornell University, 2007

15.5 PATENTS

- 2015 M. Takáč, S. D. Ahipasaoglu, P. Richtárik and N. M. Cheung
Method and system for classifying images
Patent# WO/2015/011470